Probability

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Lecture 1 – 23/01/26

§1 Fundamentals

Definition 1.1. Let Ω be a set and $\mathcal{F}$ a collection of subsets of Ω. We call $\mathcal{F}$ a *σ-algebra* if

- $\Omega \in \mathcal{F}$;
- if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$;
- if $(A_n)_{n\in\mathbb{N}}$ is a countable collection of sets in $\mathcal{F}$, then
$$\bigcup A_n \in \mathcal{F}$$

Definition 1.2. Let $\mathcal{F}$ be a σ-algebra on Ω. A function
$$\mathbb{P} : \mathcal{F} \to [0,1]$$
is called a *probability measure* if

- $\mathbb{P}(\Omega) = 1$;
- for any countable disjoint collection $(A_n)_{n\in\mathbb{N}}$ in $\mathcal{F}$,
$$\mathbb{P}\left(\bigcup A_n\right) = \sum \mathbb{P}(A_n)$$

We call $\mathbb{P}(A)$ the *probability* of A.

Then $(\Omega, \mathcal{F}, \mathbb{P})$ is a *probability space*.

Definition 1.3. The elements of Ω are called *outcomes*, and the elements of the σ-algebra $\mathcal{F}$ are called *events*.

Importantly, we talk about probabilities of *events*, never probabilities of outcomes – the domain of $\mathbb{P}$ is $\mathcal{F}$, not Ω.

> **Remark.** When Ω is countable, we take $\mathcal{F}$ to be all subsets of Ω. However, this framework will become useful when considering uncountable Ω.

Let us note some important properties of probability measures that follow from the definition.

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;
- $\mathbb{P}(\emptyset) = 0$;

- $\mathbb{P}(A) \leq \mathbb{P}(B)$ if $A \subseteq B$;
- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Example 1.4

- *Rolling a fair die*. Then $\Omega = \{1, 2, 3, 4, 5, 6\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, and
$$\mathbb{P}(A) = \frac{|A|}{6}$$
since all outcomes are equally likely.
- $\Omega = \{\omega_1, \ldots, \omega_n\}$, $\mathcal{F} = \mathcal{P}(\Omega)$, $\mathbb{P}(A) = \frac{|A|}{|\Omega|}$ models the experiment of picking an element of Ω uniformly at random.
- *Picking balls from a bag*. Suppose we have n balls labelled $\{1, \ldots, n\}$, and we pick $k \leq n$ balls uniformly at random without replacement. Then $\Omega = \{A : A \subseteq \{1, \ldots, n\}, |A| = k\}$, so $|\Omega| = \binom{n}{k}$ and $\mathbb{P}(\{\omega\}) = \binom{n}{k}^{-1}$ for any $\omega \in \Omega$.
- *A deck of 52 cards*. Take a well-shuffled deck of 52 cards. Then $\Omega = \{\text{all permutations of 52 cards}\}$, so $|\Omega| = 52!$.

  What is, for example, $\mathbb{P}(\text{top 2 cards are aces})$? It is $\frac{4 \cdot 3 \cdot 50!}{52!}$.
- *Largest digit*. Consider a string of n random digits from 0 up to 9. Then $\Omega = \{0, \ldots, 9\}^n$, $|\Omega| = 10^n$.

  Consider the events $A_k = \{\text{no digit exceeds } k\}$, $B_k = \{\text{largest digit is } k\}$. Then
$$\begin{aligned} \mathbb{P}(A_k) &= \frac{(k+1)^n}{10^n} \\ \implies \mathbb{P}(B_k) = \mathbb{P}(A_k \setminus A_{k-1}) &= \frac{(k+1)^n - k^n}{10^n} \end{aligned}$$
- *Birthday problem*. There are n people. What is the probability that at least 2 of them share the same birthday?

  Assume each birthday is equally likely to be one of $\{1, \ldots, 365\}$. Then $\Omega = \{1, \ldots, 365\}^n$, $|\Omega| = 365^n$, and $\mathbb{P}(\{\omega\}) = \frac{1}{365^n}$.

  Let $A = \{\text{at least 2 people have the same birthday}\}$. Instead we count $A^{\mathsf{c}} = \{\text{all birthdays different}\}$.
$$\begin{aligned} \mathbb{P}(A^{\mathsf{c}}) &= \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \\ \implies \mathbb{P}(A) &= 1 - \frac{365 \cdot 364 \cdots (365 - n + 1)}{365^n} \end{aligned}$$
  Perhaps surprisingly, if $n = 22$, $\mathbb{P}(A) = 0.476$, while if $n = 23$, $\mathbb{P}(A) = 0.507$.

§1.1 Combinatorial analysis

Let us introduce some notions.

1. Let Ω be a finite set with $|\Omega| = n$. We want to partition Ω into k disjoint subsets $\Omega_1, \dots, \Omega_k$ with $|\Omega_i| = n_i$ and $\sum n_i = n$. How many ways are there to do this? It is

$$\begin{aligned} M &= \binom{n}{n_1} \cdot \binom{n - n_1}{n_2} \cdots \binom{n - (n_1 + \cdots + n_{k-1})}{n_k} \\ &= \frac{n!}{n_1! n_2! \cdots n_k!} \\ &= \binom{n}{n_1, n_2, \dots, n_k} \end{aligned}$$

   This is the *multinomial coefficient.*

2. Suppose $f : \{1, \dots, k\} \to \{1, \dots, n\}$ (with $k \leq n$). f is *strictly increasing* if $x < y \implies f(x) < f(y)$, and f is *increasing* if $x < y \implies f(x) \leq f(y)$.

   How many strictly increasing $f : \{1, \dots, k\} \to \{1, \dots, n\}$ are there? Such a function is determined uniquely by its range, so there are $\binom{n}{k}$ of them.

   What about increasing functions? Let's define a bijection

$$\{f : \{1, \dots, k\} \to \{1, \dots, n\} \text{ inc.}\} \to \{f : \{1, \dots, k\} \to \{1, n + k - 1\} \text{ strictly inc.}\}$$

   which takes f to g with $g(i) = f(i) + i - 1$. Hence we have $\binom{n+k-1}{k}$ such functions.

Definition 1.5 (Asymptotic notation)**.** If (a_n) and (b_n) are sequences, we write $a_n \sim b_n$ if

$$\lim_{n \to \infty} \frac{a_n}{b_n} = 1$$

§1.2 Stirling's formula

Theorem 1.6 (Stirling)

$$n! \sim n^n \sqrt{2\pi n} e^{-n}$$

as $n \to \infty$.

Weaker statement. We shall first show a weaker statement, that $\log n! \sim n \log n$.

Define $l_n = \log n! = \log 2 + \cdots + \log n$. We have

$$\log \lfloor x \rfloor \leq \log x \leq \log \lfloor x + 1 \rfloor$$

Integrating from 1 to n gives

$$\begin{aligned} \sum_{k=1}^{n-1} \log k &\leq \int_1^n \log x \, dx \leq \sum_{k=1}^{n} \log k \\ \implies l_{n-1} &\leq n \log n - n + 1 \leq l_n \\ \implies n \log n - n + 1 &\leq l_n \leq (n+1)\log(n+1) - (n+1) + 1 \end{aligned}$$

Now the result follows by dividing through by $n \log n$ and taking the limit. ∎

Non-examinable proof of Stirling. For every twice-differentiable $f : \mathbb{R} \to \mathbb{R}$, for all $a < b$, we have

$$\int_a^b f(x)\ dx = \frac{f(a)+f(b)}{2}(b-a) - \frac{1}{2}\int_a^b (x-a)(b-x)f''(x)\ dx$$

obtained by integrating twice by parts. Put $f = \log$, $a = k$, $b = k+1$ for $k \in \mathbb{N}$, to get

$$\int_k^{k+1} \log x\ dx = \frac{\log k + \log(k+1)}{2} + \frac{1}{2}\int_k^{k+1} \frac{(x-k)(k+1-x)}{x^2}\ dx$$

Summing from $k = 1$ to $n-1$, we have

$$\int_1^n \log x\ dx = \frac{\log(n-1)! + \log n!}{2} + \sum_{k=1}^{n-1} a_k$$

where $a_k = \frac{1}{2}\int_0^1 \frac{x(1-x)}{(x+k)^2}\ dx$. Hence

$$n\log n - n + 1 = \log n! - \frac{\log n}{2} + \sum_{k=1}^{n-1} a_k$$

and so

$$\begin{aligned} \log n! &= n\log n - n + 1 + \frac{\log n}{2} - \sum_{k=1}^{n-1} a_k \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot \exp\left(1 - \sum_{k=1}^{n-1} a_k\right) \end{aligned}$$

Let us now consider

$$\begin{aligned} a_k &= \frac{1}{2}\int_0^1 \frac{x(1-x)}{(x+k)^2}\ dx \\ &\le \frac{1}{2}\int_0^1 \frac{1}{2k^2}\int_0^1 x(1-x)\ dx \\ &= \frac{1}{12k^2} \end{aligned}$$

and so $\sum_{k=1}^{\infty} a_k < \infty$. Now put

$$\begin{aligned} A &= \exp\left(1 - \sum_{k=1}^{\infty} a_k\right) \\ \implies n! &= n^n \cdot e^{-n} \cdot \sqrt{n} \cdot A \cdot \underbrace{\exp\left(\sum_{k=n}^{\infty} a_k\right)}_{\to 1 \text{ as } n \to \infty} \\ \implies \frac{n!}{n^n e^{-n}\sqrt{n}} &\to A \text{ as } n \to \infty \end{aligned}$$

It remains to show that $A = \sqrt{2\pi}$. Consider

$$2^{-2n}\binom{2n}{n} = 2^{-2n}\frac{(2n)!}{(n!)^2}$$

$$\sim \frac{2^{-2n}(2n)^{2n}e^{-2n}\sqrt{2n}A}{n^{2n}e^{-2n}nA^2}$$
$$= \frac{\sqrt{2}}{A\sqrt{n}}$$

Now, we shall use a different method to show

$$2^{-2n}\binom{2n}{n} \sim \frac{1}{\sqrt{\pi n}} \text{ as } n \to \infty$$

Consider

$$I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$$

We have $I_0 = \frac{\pi}{2}$, $I_1 = 1$, and, by integrating by parts, we have

$$I_n = \frac{n-1}{n} I_{n-2} \qquad (*)$$

so that

$$\begin{aligned} I_{2n} &= \frac{2n-1}{2n} I_{2n-2} \\ &= \cdots = \frac{(2n-1)(2n-3)\cdots 3 \cdot 1}{(2n)(2n-2)\cdots 2} \cdot \frac{\pi}{2} \\ &= \frac{2n(2n-1)(2n-2)\cdots 3 \cdot 2 \cdot 1}{2n \cdot 2n \cdot (2n-2) \cdot (2n-2) \cdots 2 \cdot 2} \cdot \frac{\pi}{2} \\ &= \frac{(2n)!}{2^{2n}(n!)^2} \cdot \frac{\pi}{2} \\ &= \frac{\pi}{2} \cdot 2^{-2n}\binom{2n}{n} \end{aligned}$$

and we can similarly obtain

$$\begin{aligned} I_{2n+1} &= \frac{2n \cdots 4 \cdot 2}{(2n+1)\cdots 3 \cdot 1} \cdot 1 \\ &= \frac{1}{2n+1}\left(2^{-2n}\binom{2n}{n}\right)^{-1} \end{aligned}$$

Now, we want to show that $\frac{I_{2n}}{I_{2n+1}} \to 1$ as $n \to \infty$. By $(*)$, we have

$$\frac{I_n}{I_{n-2}} \to 1 \text{ as } n \to \infty$$

But $I_n = \int_0^{\pi/2} \cos^n \theta \, d\theta$, so I_n is decreasing in n. This gives

$$\frac{I_{2n}}{I_{2n-2}} \le \frac{I_{2n}}{I_{2n+1}} \le \frac{I_{2n-1}}{I_{2n+1}}$$

which shows the result by sandwich. Hence, we get, as $n \to \infty$,

$$\frac{\frac{\pi}{2} \cdot 2^{-2n} \cdot \binom{2n}{n}}{\frac{1}{2n+1} \cdot \left(2^{-2n}\binom{2n}{n}\right)^{-1}} \to 1$$
$$\implies \left(2^{-2n}\binom{2n}{n}\right)^2 \sim \frac{1}{\pi n}$$

■

Proposition 1.7 (Countable subadditivity)
Let (A_n) be a collection of events in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n\right) \leq \sum \mathbb{P}(A_n)$$

Proof. Let $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. By definition, it's immediate that (B_n) is a disjoint collection of sets in $\mathcal{F}$, and $\bigcup B_n = \bigcup A_n$, so

$$\begin{aligned} \mathbb{P}\left(\bigcup A_n\right) = \mathbb{P}\left(\bigcup B_n\right) &= \sum \mathbb{P}(B_n) \\ &\leq \sum \mathbb{P}(A_n) \end{aligned}$$

since $B_n \subseteq A_n$ for all n. ∎

Proposition 1.8 (Continuity of probability measures)
Take a sequence of events $(A_n)_{n\in\mathbb{N}}$ such that $A_n \subseteq A_{n+1}$ for all n – we say (A_n) is an *increasing* sequence.

Then $\mathbb{P}(A_n)$ is increasing and converges. In particular,

$$\mathbb{P}(A_n) \to \mathbb{P}\left(\bigcup A_n\right)$$

as $n \to \infty$.

Proof. Define $B_1 = A_1$, and for $n \geq 2$, let $B_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$. Then

$$\bigcup_{k=1}^{n} B_k = A_n$$

and the B_n's are disjoint. Then

$$\begin{aligned} \mathbb{P}(A_n) &= \mathbb{P}\left(\bigcup_{k=1}^{n} B_k\right) \\ &= \sum_{k=1}^{n} \mathbb{P}(B_k) \to \sum_{k=1}^{\infty} \mathbb{P}(B_k) \end{aligned}$$

Also,

$$\begin{aligned} \sum_{k=1}^{\infty} \mathbb{P}(B_k) &= \mathbb{P}\left(\bigcup B_k\right) \\ &= \mathbb{P}\left(A_k\right) \end{aligned}$$

∎

Corollary 1.9

Analogously, take a sequence of events (A_n) such that $A_{n+1} \subseteq A_n$ for all n – we say (A_n) is a decreasing sequence. Then

$$\mathbb{P}(A_n) \to \mathbb{P}\left(\bigcap A_n\right)$$

Proof. Take complements and use the previous claim. ■

Proposition 1.10 (Inclusion-exclusion formula for probability measures)

Let $A_1, \dots, A_n \in \mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_i\right) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

Proof. We proceed by induction.

Base case. ADD

Inductive step. We have

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) &= \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cup A_n\right) \\
&= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\left(\bigcup_{i=1}^{n-1} A_i\right) \cap A_n\right) \\
&= \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right)
\end{aligned}$$

Applying the inductive hypothesis, we get

$$\begin{aligned}
\mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) \\
\mathbb{P}\left(\bigcup_{i=1}^{n-1} B_i\right) &= \sum_{k=1}^{n-1} (-1)^{k+1} \sum_{1 \leq i_1 < \cdots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k} \cap A_n)
\end{aligned}$$

and subbing this in, we will find that everything works out. ■

Proposition 1.11 (Bonferroni inequalities)
Let $A_1, \dots, A_n$ be events. Then

$$\mathbb{P}\left(\bigcup A_i\right) \begin{cases} \leq \sum_{k=1}^{r} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ odd} \\ \geq \sum_{k=1}^{r} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) & \text{if } r \text{ even} \end{cases}$$

where those sums are just truncated expansions of inclusion-exclusion.

Proof. We induct again.
Base case. $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B) \leq \mathbb{P}(A) + \mathbb{P}(B)$
Inductive step. Suppose r is odd. Then

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) = \mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) + \mathbb{P}(A_n) - \mathbb{P}\left(\bigcup_{i=1}^{n-1} \underbrace{(A_i \cap A_n)}_{B_i}\right)$$

By the inductive hypotheses,

$$\mathbb{P}\left(\bigcup_{i=1}^{n-1} A_i\right) \leq \sum_{k=1}^{r} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})$$
$$\mathbb{P}\left(\bigcup_{i=1}^{n-1} (A_i \cap A_n)\right) \geq \sum_{k=1}^{r-1} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n-1} \mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k} \cap A_n)$$

where in the former, we choose to apply the identity for odd r, and in the latter, we apply it for even $r - 1$. Substituting much as for the proof of the inclusion-exclusion formula, we obtain the result. The even case is analogous. ■

Let us now consider the special case where Ω is a finite set and $\mathbb{P}(A) = |A|/|\Omega|$. Then we have

Corollary 1.12 (Inclusion-exclusion principle for sets)
Suppose Ω is finite. Then

$$|A_1 \cup \dots \cup A_n| = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_n}|$$

Proof. Consider the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ where $\mathbb{P}(A) = |A|/|\Omega|$. Then, apply inclusion-exclusion and cancel $1/|\Omega|$ from both sides. ■

§2 Basic applications

§2.1 Counting surjections

Define a probability space as follows:

$$\Omega = \{f : \{1, \dots, n\} \to \{1, \dots, m\}\}$$

$$A = \{f \in \Omega : f \text{ is a surjection}\}$$

What is $|A|$? Let us define a set of events

$$A_i = \{f \in \Omega : i \notin \{f(1), \dots, f(n)\}\}, \, i \in \{1, \dots, m\}$$

so that

$$\begin{aligned}
A &= A_1^c \cap \dots \cap A_m^c = (A_1 \cup \dots \cup A_m)^c \\
\implies |A| &= |\Omega| - |A_1 \cup \dots \cup A_m| \\
&= |\Omega| - \sum_{k=1}^{m} (-1)^{k+1} \sum_{1 \le i_1 < \dots < i_k \le m} |A_{i_1} \cap \dots \cap A_{i_k}| \\
&= \underbrace{|\Omega|}_{m^n} - \sum_{k=1}^{m} (-1)^{k+1} \binom{m}{k} (m-k)^n \\
&= \sum_{k=0}^{m} (-1)^k \binom{m}{k} (m-k)^n
\end{aligned}$$

Lecture 4 – 30/01/26

§2.2 Counting derangements

Definition 2.1. A *derangement* is a permutation with no fixed points.

Let

$$\begin{aligned}
\Omega &= \{\text{permutations of } \{1, \dots, n\}\} \\
A &= \{\text{derangements}\}
\end{aligned}$$

Let

$$A_i = \{f \in \Omega : f(i) = i\}$$

so that

$$\begin{aligned}
A &= A_1^c \cap \dots \cap A_n^c = (A_1 \cup \dots \cup A_n)^c \\
\implies \mathbb{P}(A) = 1 - \mathbb{P}\left(\bigcup A_i\right) &= 1 - \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \le i_1 < \dots < i_k \le n} \underbrace{\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k})}_{\frac{(n-k)!}{n!}} \\
&= 1 - \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} \frac{(n-k)!}{n!} \\
&= 1 - \sum_{k=1}^{n} (-1)^{k+1} \frac{1}{k!} \\
&= \sum_{k=0}^{n} \frac{(-1)^k}{k!} \to e^{-1}
\end{aligned}$$

§3 Independence and conditional probability

Definition 3.1 (Independence)**.** Let A, $B \in \mathcal{F}$. We say A and B are *independent* and write $A \perp B$ if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Likewise, a countable collection of events $(A_n)_{n \in \mathbb{N}}$ is said to be independent if, for distinct $i_1, \ldots, i_k$,

$$\mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k}) = \prod_{j=1}^{k} \mathbb{P}(A_{i_j})$$

> **Remark.** Pairwise independence of events does not imply independence.

Example

Toss a fair coin twice. We have

$$\begin{aligned} \Omega &= \{(0,0), (0,1), (1,0), (1,1)\} \\ \mathbb{P}(\{\omega\}) &= \frac{1}{4} \end{aligned}$$

Let

$$\begin{aligned} A &= \{(0,0), (0,1)\} \\ B &= \{(0,0), (1,0)\} \\ C &= \{(1,0), (0,1)\} \end{aligned}$$

We have $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C)$, and furthermore we can check that $A \perp B$, $A \perp C$ and $B \perp C$. But
$P(A \cap B \cap C) = \mathbb{P}(\emptyset) = 0$, so they are not independent.

Proposition 3.2

If A is independent of B, then A is independent of B^{c}.

Proof. We have

$$\begin{aligned} \mathbb{P}(A \cap B^{\mathsf{c}}) &= \mathbb{P}(A) - \mathbb{P}(A \cap B) \\ &= \mathbb{P}(A)(1 - \mathbb{P}(B)) \\ &= \mathbb{P}(A)\mathbb{P}(B^{\mathsf{c}}) \end{aligned}$$

∎

Definition 3.3 (Conditional probability)**.** Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $B \in \mathcal{F}$ with $\mathbb{P}(B) > 0$. For $A \in \mathcal{F}$ we write the *conditional probability of A given B*

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$$

Remark. Notably, if $A \perp B$, then $\mathbb{P}(A|B) = \mathbb{P}(A)$, as we would expect.

Proposition 3.4 (Countable additivity for conditional probability)
Suppose $(A_n)_{n\in\mathbb{N}}$ is a disjoint collection in $\mathcal{F}$. Then

$$\mathbb{P}\left(\bigcup A_n|B\right) = \sum \mathbb{P}(A_n|B)$$

Proof. We have

$$\begin{aligned}
\mathbb{P}\left(\bigcup A_n|B\right) &= \frac{\mathbb{P}\left(\bigcup A_n \cap B\right)}{\mathbb{P}(B)} \\
&= \frac{\mathbb{P}\left(\bigcup (A_n \cap B)\right)}{\mathbb{P}(B)} \\
&= \sum \frac{\mathbb{P}(A_n \cap B)}{\mathbb{P}(B)} \\
&= \sum \mathbb{P}(A_n|B)
\end{aligned}$$

■

Proposition 3.5 (Law of total probability)
Suppose $(B_n)_{n\in\mathbb{N}}$ is a disjoint collection in $\mathcal{F}$, $\bigcup B_n = \Omega$, and $\mathbb{P}(B_n) > 0$ for every n. Let $A \in \mathcal{F}$, then

$$\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$$

Proof. We have

$$\begin{aligned}
\mathbb{P}(A) &= \mathbb{P}\left(A \cap \left(\bigcup B_n\right)\right) \\
&= \mathbb{P}\left(\bigcup (B_n \cap A)\right) \\
&= \sum \mathbb{P}(B_n \cap A) \\
&= \sum \mathbb{P}(A|B_n)\mathbb{P}(B_n)
\end{aligned}$$

■

Proposition 3.6 (Bayes' theorem)
Suppose (B_n) are disjoint, $\bigcup B_n = \Omega$ and $P(B_n) > 0$ for every n. Let $A \in \mathcal{F}$, then

$$\mathbb{P}(B_n|A) = \frac{\mathbb{P}(A|B_n) \cdot \mathbb{P}(B_n)}{\sum_k \mathbb{P}(A|B_k)\mathbb{P}(B_k)}$$

Proof sketch. Use the definitions of $\mathbb{P}(B_n|A)$ and $\mathbb{P}(A|B_n)$, and expand the bottom using the law of total probability. ∎

This formula is the basis of so-called Bayesian statistics. We know the probability of $\mathbb{P}(B_n)$ and we have a model which gives us $\mathbb{P}(A|B_n)$. With the formula, we find the *posterior* probability of B_n given that A occurred.

Example (False positives for a rare condition)

Suppose that condition A affects 0.1% of the population. We have a medical test which is positive for 98% of the affected population and 1% of the unaffected population. What is the probability that a random individual who tests positive has the disease?

Put $A = \{\text{individual suffers from } A\}$, $P = \{\text{individual tests positive}\}$. We have $\mathbb{P}(A) = 0.001$, $\mathbb{P}(P|A) = 0.98$, $\mathbb{P}(P|A^\mathsf{c}) = 0.01$, so

$$\begin{aligned}\mathbb{P}(A|P) &= \frac{\mathbb{P}(P|A)\mathbb{P}(A)}{\mathbb{P}(P|A)\mathbb{P}(A) + \mathbb{P}(P|A^\mathsf{c})\mathbb{P}(A^\mathsf{c})} \\ &= 0.089\end{aligned}$$

In particular, we can see

$$\mathbb{P}(A|P) = \frac{1}{1 + \frac{\mathbb{P}(P|A^\mathsf{c})\mathbb{P}(A^\mathsf{c})}{\mathbb{P}(P|A)\mathbb{P}(A)}}$$

Since $\mathbb{P}(A^\mathsf{c})$ and $\mathbb{P}(P|A)$ are both very close to one, the relevant ratio is

$$\frac{\mathbb{P}(P|A^\mathsf{c})}{\mathbb{P}(A)}$$

But since $\mathbb{P}(P|A^\mathsf{c}) \gg \mathbb{P}(A)$, this causes a very small value for $\mathbb{P}(A|P)$.

Lecture n

Example
- What is the probability that I have two boys given that I have two children one of whom is a boy?

  Since no further information is given, we take all outcomes to be equally likely. Let

  $$\mathrm{BG} = \{\text{elder child a boy, younger a girl}\}$$

  with GB, BB, GG defined analogously. Then we want

  $$\begin{aligned}\mathbb{P}(\mathrm{BB}|\mathrm{BB}\cup\mathrm{BG}\cup\mathrm{GB}) &= \frac{\mathbb{P}(\mathrm{BB})}{\mathbb{P}(\mathrm{BB}\cup\mathrm{BG}\cup\mathrm{GB})}\\ &= \frac{\frac{1}{4}}{\frac{3}{4}} = \frac{1}{3}\end{aligned}$$

- What about $\mathbb{P}(\text{both boys}|\text{eldest a boy})$? This is

  $$\mathbb{P}(\mathrm{BB}|\mathrm{BB}\cup\mathrm{BG}) = \frac{1}{2}$$

- What about $\mathbb{P}(\text{both boys}|\text{one a boy born on a Thursday})$? Denote a boy born on Thursday by T, a boy not born on Thursday by N, and a girl by G. Then we want

  $$\begin{aligned}&\mathbb{P}(\mathrm{TT}\cup\mathrm{TN}\cup\mathrm{NT}|\mathrm{TT}\cup\mathrm{TG}\cup\mathrm{GT}\cup\mathrm{TN}\cup\mathrm{NT})\\ &= \frac{\mathbb{P}(\mathrm{TT}\cup\mathrm{TN}\cup\mathrm{NT})}{\mathbb{P}(\mathrm{TT}\cup\mathrm{TG}\cup\mathrm{GT}\cup\mathrm{TN}\cup\mathrm{NT})}\\ &= \frac{\left(\frac{1}{2}\cdot\frac{1}{7}\right)^2 + \left(\frac{1}{2}\cdot\frac{1}{7}\right)\left(\frac{1}{2}\cdot\frac{6}{7}\right) + \left(\frac{1}{2}\cdot\frac{6}{7}\right)\left(\frac{1}{2}\cdot\frac{1}{7}\right)}{\left(\frac{1}{2}\cdot\frac{1}{7}\right)^2 + \left(\frac{1}{2}\cdot\frac{1}{7}\right)\cdot\frac{1}{2} + \frac{1}{2}\cdot\left(\frac{1}{2}\cdot\frac{1}{7}\right) + \left(\frac{1}{2}\cdot\frac{1}{7}\right)\left(\frac{1}{2}\cdot\frac{6}{7}\right) + \left(\frac{1}{2}\cdot\frac{6}{7}\right)\left(\frac{1}{2}\cdot\frac{1}{7}\right)}\\ &= \frac{13}{27}\end{aligned}$$

  Morally speaking, we're providing information that helps specify the boy in question more, which brings the probability closer to a half.

Example (Simpson's paradox)

Consider the following data:

All applicants	Admitted	Rejected	% Admitted
State	25	25	50%
Indep.	28	22	56%

London schools	Admitted	Rejected	% Admitted
State	15	22	41%
Indep.	5	8	38%

Cambridge schools	Admitted	Rejected	% Admitted
State	10	3	77%
Indep.	23	14	62%

– paradoxically, the state admission rate is higher when the school locations are taken individually, but the independent admission right is higher for the combined data. This phenomenon is called *confounding*, and arises when we aggregate data from disparate populations.

Let

$$\begin{aligned} A &= \{\text{individual is admitted}\} \\ B &= \{\text{individual from London}\} \\ C &= \{\text{individual from state school}\} \end{aligned}$$

Then

$$\begin{aligned} \mathbb{P}(A|B \cap C) &> \mathbb{P}(A|B \cap C^{\mathsf{c}}) \\ \mathbb{P}(A|B^{\mathsf{c}} \cap C) &> \mathbb{P}(A|B^{\mathsf{c}} \cap C^{\mathsf{c}}) \\ \text{but } \mathbb{P}(A|C) &< \mathbb{P}(A|C^{\mathsf{c}}) \end{aligned}$$

Let's examine how this can happen. We have

$$\begin{aligned} \mathbb{P}(A|C) &= \mathbb{P}(A \cap B|C) + \mathbb{P}(A \cap B^{\mathsf{c}}|C) \\ &= \frac{\mathbb{P}(A \cap B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A \cap B^{\mathsf{c}} \cap C)}{\mathbb{P}(C)} \\ &= \frac{\mathbb{P}(A|B \cap C)\mathbb{P}(B \cap C)}{\mathbb{P}(C)} + \frac{\mathbb{P}(A|B^{\mathsf{c}} \cap C)\mathbb{P}(B^{\mathsf{c}} \cap C)}{\mathbb{P}(C)} \\ &= \mathbb{P}(A|B \cap C)\mathbb{P}(B|C) + \mathbb{P}(A|B^{\mathsf{c}} \cap C^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}}|C^{\mathsf{c}}) \\ &> P(A|B \cap C^{\mathsf{c}})\mathbb{P}(B|C) + \mathbb{P}(A|B^{\mathsf{c}} \cap C^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}}|C) \end{aligned}$$

by the first two inequalities from above. Now, if $\mathbb{P}(B|C) = \mathbb{P}(B|C^{\mathsf{c}})$, then we would get $\mathbb{P}(A|C) > \mathbb{P}(A|C^{\mathsf{c}})$, which is the intuitive expectation – but here this does not hold, and so we can make the 'paradox' hold.

Even more simply, this is just the fact that, if $\frac{a_1}{a_2} > \frac{b_1}{b_2}$ and $\frac{c_1}{c_2} > \frac{d_1}{d_2}$, it does not necessarily hold that $\frac{a_1+c_1}{a_2+c_2} > \frac{b_1+d_1}{b_2+d_2}$.

§4 Probability distributions

§4.1 Discrete probability distributions

Take a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with Ω either finite or countable – that is, $\Omega = \{\omega_1, \omega_2, \dots\}$. As normal, we take $\mathcal{F} = \mathcal{P}(\Omega)$, and so we need only care about $\mathbb{P}(\{\omega_i\})$ for every i.

Definition 4.1 (Discrete probability distributions)**.** For such a setup, we call the sequence of probabilities $(\mathbb{P}(\{\omega_i\}))_i$ a *discrete probability distribution*. We may also write $p_i = \mathbb{P}(\{\omega_i\})$.

Let us consider some examples of discrete probability distributions.

① *Bernoulli distribution.* The Bernoulli distribution models the outcome of a toss of a biased coin. We have $\Omega = \{0, 1\}$, $p_0 = \mathbb{P}(\{0\}) = 1 - p$, $p_1 = \mathbb{P}(\{1\}) = p$ for some p with $p \in [0, 1]$.

② *Binomial distribution.* We denote this $\mathrm{Bin}(n, p)$, with $n \in \mathbb{N}$, $p \in [0, 1]$. It models the number of heads seen for n independent tosses of a p-biased coin. We have

$$\begin{aligned} \Omega &= \{0, 1, \dots, n\} \\ p_k &= \binom{n}{k} p^k (1-p)^{n-k} \end{aligned}$$

③ *Multinomial distribution.* We denote this $\mathrm{M}(n, p_1, \dots, p_k)$, where $p_i \geq 0$, $\sum p_i = 1$. Given k boxes and n balls placed in box i with p_i, the multinomial distribution gives the probability of n_l balls in box l for every n. We have

$$\begin{aligned} \Omega &= \left\{ (n_1, \dots, n_k) \in \mathbb{N}^k : \sum_{i=1}^k n_i = n \right\} \\ \mathbb{P}(n_1, \dots, n_k) &= \binom{n}{n_1} n_1^{p_1} \cdot \binom{n - n_1}{n_2} p_2^{n_2} \cdots \binom{n - n_1 - \dots - n_{k-1}}{n_k} p_k^{n_k} \\ &= \binom{n}{n_1, n_2, \dots, n_k} p_1^{n_1} p_2^{n_2} \dots p_k^{n_k} \end{aligned}$$

④ *Geometric distribution.* The geometric distribution gives the probabiity that the first heads achieved on a p-coin is after k tosses:

$$\begin{aligned} \Omega &= \{1, 2, \dots\} \\ p_k &= (1-p)^{k-1} p \end{aligned}$$

Alternatively, we might want to model the probability that we see k tails before the first heads:

$$\begin{aligned} \Omega &= \{0, 1, \dots\} \\ p'_k &= (1-p)^k p \end{aligned}$$

⑤ *Poisson distribution.* Suppose customers arrive into a shop in $[0, 1]$. Discretise $[0, 1]$ into $\left\{ \left[\frac{i-1}{N}, \frac{i}{N} \right] \right\}, i = 1, \dots, N$, $N \in \mathbb{N}$. Independently, in each interval, a customer arrives with probability p, and doesn't arrive with probability $1 - p$.

Then

$$\mathbb{P}(k \text{ customers have arrived}) = \binom{N}{k} p^k (1-p)^{N-k}$$

– in other words, this follows a binomial distribution. Now, take $p = \frac{\lambda}{N}$, so

$$\begin{aligned} \mathbb{P}(k \text{ customers have arrived}) &= \frac{N!}{k!(N-k)!} \left(\frac{\lambda}{N}\right)^k \left(1 - \frac{\lambda}{N}\right)^{N-k} \\ &= \frac{\lambda^k}{k!} \cdot \underbrace{\frac{N(N-1)\dots(N-k+1)}{N^k}}_{\to 1} \underbrace{\left(1 - \frac{\lambda}{N}\right)^{N-k}}_{\to e^{-\lambda}} \end{aligned}$$

$$\to e^{-\lambda}\frac{\lambda^k}{k!}$$

as $N \to \infty$. Hence we have

$$\begin{aligned}\Omega &= \{0, 1, 2, \dots\} \\ p_k &= e^{-\lambda}\frac{\lambda^k}{k!}\end{aligned}$$

and we denote this $\mathrm{Poi}(\lambda)$.

§4.2 Random variables

Definition 4.2. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \to \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \le x\} \in \mathcal{F}$$

for any x. We have the shorthand notation $\{X \le x\}$ to refer to $\{\omega \in \Omega : X(\omega) \le x\}$, and $\{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\}$.

Definition 4.3. Let $A \in \mathcal{F}$. Define the *indicator* of A

$$1_A(\omega) = 1(\omega \in A) = \begin{cases} 1 & \omega \in A \\ 0 & \omega \notin A \end{cases}$$

Since $A \in \mathcal{F}$, 1_A is a random variable.

Definition 4.4. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned} F_X(x) : \mathbb{R} &\to [0, 1] \\ x &\mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \le x\}) = \mathbb{P}(X \le x) \end{aligned}$$

Definition 4.5. $X = (X_1, \dots, X_n)$ is called a *random variable in* $\mathbb{R}^n$ if

$$(X_1, \dots, X_n) : \Omega \to \mathbb{R}^n$$

is a function such that, for any $x_1, \dots, x_n$,

$$\{\omega : X_1(\omega) \le x_1, \dots, X_n(\omega) \le x_n\} \in \mathcal{F}$$

which we denote with the shorthand $\{X_1 \le x_1, \dots, X_n \le x_n\}$.

Note that $(X_1, \dots, X_n)$ is a random variable if and only if all of $X_1, \dots, X_n$ are.

Definition 4.6. A random variable X is called *discrete* if it takes values in some countable set S.

Definition 4.7. For a discrete random variable, we are able to write

$$p_x = \mathbb{P}(X = x) = \mathbb{P}(\omega : X(\omega) = x)$$

Then, $(p_x)_{x \in S}$ is called the *probability mass function* of X.

Now we are able to say things like

$$\text{for any } A \subseteq S, \mathbb{P}(X \in A) = \sum_{x \in A} p_x$$

If p_x follows a given distribution, then we say X is a random variable with that distribution or a random variable following that distribution.

Definition 4.8. Let $X_1, \dots, X_n$ be discrete random variables with values in $S_1, \dots, S_n$. They are called *independent* if

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \mathbb{P}(X_1 = x_1) \dots \mathbb{P}(X_n = x_n)$$

for all $x_1 \in S_1, \dots, x_n \in S_n$.

Example 4.9

Toss a p-coin N times independently. Then

$$\Omega = \{0,1\}^N$$

and every $\omega \in \Omega$ is of the form $(\omega_1, \dots, \omega_N)$, where ω_i is the i^{th} outcome. Hence

$$p_\omega = \mathbb{P}(\{\omega\}) = \prod_{k=1}^{N} p^{\omega_k}(1-p)^{1-\omega_k}$$

Let us also define random variables X_k for each k, satisfying $X_k(\omega) = \omega_k$. Clearly X_k is a Bernoulli r.v. of parameter p. We claim $X_1, \dots, X_N$ are independent: we have

$$\begin{aligned} P(X_1 = x_1, \dots, X_N = x_N) &= \prod_{k=1}^{N} p^{x_\omega}(1-p)^{1-x_\omega} \\ &= \prod_{k=1}^{N} \mathbb{P}(X_i = x_i) \end{aligned}$$

Lecture 7 – 06/02/26

§4.3 Expectation

Definition 4.10. Suppose Ω is finite or countable, and let $X : \Omega \to \mathbb{R}$ be a discrete random variable. We say X is *non-negative* if $X \geq 0$ – then we define the *expectation* of X by

$$\mathbb{E}[X] = \sum_{\omega \in \Omega} \mathbb{P}(\{\omega\}) \cdot X(\omega)$$

We can equivalently let

$$\Omega_X = \{X(\omega) : \omega \in \Omega\}$$

so that

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{\omega\in\Omega} X(\omega)\cdot\mathbb{P}(\{\omega\}) \\
&= \sum_{x\in\Omega_X}\sum_{\omega\in\{X=x\}} X(\omega)\mathbb{P}(\{\omega\}) \\
&= \sum_{x\in\Omega_X}\sum_{\omega\in\{X=x\}} x\mathbb{P}(\{\omega\}) \\
&= \sum_{x\in\Omega_X} x \sum_{\omega\in\{X=x\}} \mathbb{P}(\{\omega\}) \\
&= \sum_{x\in\Omega_X} x\cdot\mathbb{P}(X=x)
\end{aligned}$$

or in other words the average of the values taken by X, weighted by their probabilities.

Example 4.11

Suppose $X \sim \mathrm{Bin}(n,p)$. Then

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{k=0}^{N} k\mathbb{P}(X=k) \\
&= \sum_{k=0}^{N} k\binom{N}{k}p^k(1-p)^{N-k} \\
&= \sum_{k=0}^{N} k\frac{N!}{(N-k)!k!}p^k(1-p)^{N-k} \\
&= \sum_{k=0}^{N-1} N\cdot\binom{N-1}{k}p^{k-1}(1-p)^{N-k}\cdot p \\
&= Np\cdot(p+1-p)^{N-1} = Np
\end{aligned}$$

Example 4.12

Suppose $X \sim \mathrm{Poi}(\lambda)$. Then

$$\begin{aligned}
\mathbb{E}[X] &= \sum_{k=0}^{\infty} k\mathbb{P}(X=k) \\
&= \sum_{k=0}^{\infty} ke^{-\lambda}\frac{\lambda^k}{k!} \\
&= e^{-\lambda}\cdot\lambda\sum_{k=0}^{\infty}\frac{\lambda^{k-1}}{(k-1)!} = \lambda
\end{aligned}$$

Let X be a random variable – then we define

$$\begin{aligned}
X^+ &= \max(X,0) \\
X^- &= \max(-X,0)
\end{aligned}$$

so $X = X^+ - X^-$ and $|X| = X^+ + X^-$. Supposing X is discrete, we can define the expectations of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$, since they are both non-negative.

If at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite, we define

$$\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$$

If both are infinite, the expectation of X is undefined.

Also, if $\mathbb{E}[|X|]$ (which we define in the obvious way) is $< \infty$, we say X is integrable. When $\mathbb{E}[X]$ is well-defined, we have

$$\mathbb{E}[X] = \sum_{x \in \Omega_X} x \cdot \mathbb{P}(X = x)$$

We have some properties of expectations as follows:

- If $X \geq 0$, then $\mathbb{E}[X] \geq 0$.
- If $X \geq 0$ and $\mathbb{E}[X] = 0$, then $\mathbb{P}(X = 0) = 1$.
- If $c \in \mathbb{R}$, then $\mathbb{E}[cX] = c\mathbb{E}[X]$ and $\mathbb{E}[c + X] = c + \mathbb{E}[X]$.
- If X and Y are random variables (and X, Y integrable), then $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.
- In the countable case, we take $c_1, \dots, c_n \in \mathbb{R}$ and $X_1, \dots, X_n$ integrable, and then we have

$$\mathbb{E}\left[\sum_{i=1}^{n} c_i X_i\right] = \sum_{i=1}^{n} c_i \mathbb{E}[X_i]$$

Let us give a proof of the final identity in the case where Ω is countable. We have

$$\begin{aligned}
\mathbb{E}\left[\sum_n X_n\right] &= \sum_{\omega \in \Omega} \left(\sum_n X_n(\omega)\right) \mathbb{P}(\{\omega\}) \\
&= \sum_n \sum_{\omega \in \Omega} X_n(\omega) \mathbb{P}(\{\omega\}) \\
&= \sum_n \mathbb{E}[X_n]
\end{aligned}$$

where we can interchange sums by the assumption that the random variables are integrable.

We also have the useful result that, if $X = 1(A)$ for some event A,

$$\mathbb{E}[X] = \mathbb{P}(A)$$

Proposition 4.13

Let $g : \mathbb{R} \to \mathbb{R}$, and define the random variable $g(X)$ by

$$g(X)(\omega) = g(X(\omega))$$

Then

$$\mathbb{E}[g(X)] = \sum_{x \in \Omega_X} g(x) \mathbb{P}(X = x)$$

Proof. Put $Y = g(X)$, then

$$\begin{aligned}\{Y = y\} &= \{\omega : Y(\omega) = y\} \\ &= \{\omega : g(X(\omega)) = y\} \\ &= \{\omega : X(\omega) \in g^{-1}(\{y\})\} \\ &= \{X \in g^{-1}(\{y\})\}\end{aligned}$$

Hence

$$\begin{aligned}\mathbb{E}[Y] &= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(Y = y) \\ &= \sum_{y \in \Omega_Y} y \cdot \mathbb{P}(X \in g^{-1}(\{y\})) \\ &= \sum_{y \in \Omega_Y} y \sum_{x \in g^{-1}(\{y\})} \mathbb{P}(X = x) \\ &= \sum_{y \in \Omega_Y} \sum_{x \in g^{-1}(\{y\})} g(x)\mathbb{P}(X = x) \\ &= \sum_{x \in \Omega_X} g(x) \cdot \mathbb{P}(X = x)\end{aligned}$$

■

Proposition 4.14

Suppose that $X \geq 0$ and takes integer values. Then

$$\mathbb{E}[X] = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) = \sum_{k=0}^{\infty} \mathbb{P}(X > k)$$

Proof. Since $X \geq 0$ and $X \in \mathbb{N}$, we have

$$X = \sum_{k=1}^{\infty} 1(X \geq k) = \sum_{k=0}^{\infty} 1(X > k)$$

Taking expectations, we obtain

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}\left[\sum_{k=1}^{\infty} 1(X \geq k)\right] & &= \mathbb{E}\left[\sum_{k=0}^{\infty} 1(X > k)\right] \\ &= \sum_{k=1}^{\infty} \mathbb{E}[1(X \geq k)] & &= \sum_{k=0}^{\infty} \mathbb{E}[1(X > k)] \\ &= \sum_{k=1}^{\infty} \mathbb{P}(X \geq k) & &= \sum_{k=0}^{\infty} \mathbb{P}(X > k)\end{aligned}$$

■

Lecture $n + 1$ – 09/02/26

We present another proof of the inclusion-exclusion formula. First, we should establish some properties about indicator functions:

- $1(A^c) = 1 - 1(A)$;
- $1(A \cap B) = 1(A) \cdot 1(B)$;
- $1(A \cup B) = 1 - (1 - 1(A))(1 - 1(B))$.

More generally,

$$\begin{aligned}1(A_1 \cup \cdots \cup A_n) &= 1 - \prod_{i=1}^{n}(1 - 1(A_i)) \\ &= \sum_{i=1}^{n} 1(A_i) - \sum_{1 \le i_1 < i_2 \le n} 1(A_{i_1} \cap A_{i_2}) + \cdots + (-1)^{n+1} 1(A_1 \cap \cdots \cap A_n)\end{aligned}$$

Then, we can take expectations of both sides and apply $\mathbb{E}[1(A)] = \mathbb{P}(A)$ to get

$$\mathbb{P}(A_1 \cup \cdots \cup A_n) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{1 \le i_1 < \cdots < i_k \le n} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

§4.4 Moments, variance and covariance

Definition 4.15. Let X be a random variable and $r \in \mathbb{N}$. We call $\mathbb{E}[X^r]$, as long as it is well-defined, the r^{th} *moment* of X.

Definition 4.16. The *variance* of X is given by

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Furthermore, the *standard deviation* of X is given by $\sqrt{\mathrm{Var}(X)}$.

We have some properties of variance as follows:

- $\mathrm{Var}(X) \ge 0$, since it is the expectation of a squared quantity;
- If $\mathrm{Var}(X) = 0$, then $\mathbb{P}(X = \mathbb{E}[X]) = 1$;
- If $c \in \mathbb{R}$, then $\mathrm{Var}(cX) = c^2 \mathrm{Var}(X)$ and $\mathrm{Var}(X + c) = \mathrm{Var}(X)$.

Lemma 4.17

The variance of X is also given by

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$$

Proof. We have

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 \\ &= \mathbb{E}[X^2] - \mathbb{E}[X]^2\end{aligned}$$

■

Lemma 4.18

The function $\mathbb{E}[(X-c)^2]$ is minimised when $c = \mathbb{E}[X]$, and at that point is equal to $\mathrm{Var}(X)$.

Proof. We have

$$\begin{aligned} f(c) &= \mathbb{E}[(X-c)^2] \\ &= \mathbb{E}[X^2] - 2c\mathbb{E}[X] + c^2 \\ \implies f'(c) &= -2\mathbb{E}[X] + 2c = 0 \\ \implies c &= \mathbb{E}[X] \end{aligned}$$

■

Example 4.19 ① Suppose $X \sim \mathrm{Bin}(n,p)$. Recall $\mathbb{E}[X] = np$. We have

$$\mathrm{Var}(X) = \mathbb{E}[X(X-1)] + \mathbb{E}[X] - \mathbb{E}[X]^2$$

Then

$$\begin{aligned} \mathbb{E}[X(X-1)] &= \sum_{k=2}^{n} k(k-1)\frac{n!}{k!(n-k)!}p^k(1-p)^{n-k} \\ &= (n-1)np^2 \sum_{k=0}^{n-2} \binom{n-2}{k} p^k (1-p)^{n-2-k} \\ &= (n-1)np^2 \end{aligned}$$

so that $\mathrm{Var}(X) = (n-1)np^2 + np - n^2p^2 = np(1-p)$.

② Suppose $X \sim \mathrm{Poi}(\lambda)$. Recall $\mathbb{E}[X] = \lambda$. We have

$$\begin{aligned} \mathbb{E}[X(X-1)] &= \sum_{k=2}^{\infty} k(k-1)e^{-\lambda}\frac{\lambda^k}{k!} \\ &= e^{-\lambda} l^2 \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 \end{aligned}$$

So $\mathrm{Var}(X) = \lambda^2 + \lambda - \lambda^2 = \lambda$.

Definition 4.20. Let X and Y be random variables. Then the *covariance* of X and Y is

$$\mathrm{Cov}(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

We have some properties:

- $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$;
- $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$;
- for any $c \in \mathbb{R}$, $\mathrm{Cov}(cX,Y) = c\,\mathrm{Cov}(X,Y)$ and $\mathrm{Cov}(X+c,Y) = \mathrm{Cov}(X,Y)$;
- for any $c \in \mathbb{R}$, $\mathrm{Cov}(c,X) = 0$.

Lemma 4.21

The covariance of X and Y is given by

$$\operatorname{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$$

Proof. Expand. ∎

Lemma 4.22

For random variables X and Y, we have

$$\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X, Y)$$

Proof. We have

$$\begin{aligned} \operatorname{Var}(X + Y) &= \mathbb{E}[((X - \mathbb{E}[X]) + (Y - \mathbb{E}[Y]))^2] \\ &= \operatorname{Var}(X) + 2\operatorname{Cov}(X, Y) + \operatorname{Var}(Y) \end{aligned}$$

∎

Lemma 4.23

For $c_1, \ldots, c_n$ and $d_1, \ldots, d_n \in \mathbb{R}$,

$$\operatorname{Cov}\left(\sum_{i=1}^{n} c_i X_i, \sum_{j=1}^{n} d_i Y_i\right) = \sum_{i,j=1}^{n} c_i d_j \operatorname{Cov}(X_i, Y_j)$$

Corollary 4.24

We have

$$\begin{aligned} \operatorname{Var}\left(\sum_{i=1}^{n} X_i\right) &= \operatorname{Cov}\left(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_i\right) \\ &= \sum_{i=1}^{n} \operatorname{Var}(X_i) + \sum_{i \neq j} \operatorname{Cov}(X_i, X_j) \end{aligned}$$

Let us consider some results on independence.

Lemma 4.25

Suppose X_1, X_2 and X_3 are independent – then X_1 and X_2 are independent.

Proof. We have

$$\mathbb{P}(X_1 = x_1, X_2 = x_2) = \sum_{x_3} \mathbb{P}(X_1 = x_1, X_2 = x_2, X_3 = x_3)$$

$$= \sum_{x_3} \mathbb{P}(X_1 = x_1)\mathbb{P}(X_2 = x_2)\mathbb{P}(X_3 = x_3)$$
$$= \mathbb{P}(X_1 = x_1)\mathbb{P}(X_2 = x_2)$$

■

> **Remark.** The result obviously generalises to any subset of a set of independent events being independent.

Lemma 4.26

Let X and Y be two independent random variables, and let $f,\, g : \mathbb{R} \to \mathbb{R}_+$. Then

$$\mathbb{E}[f(X)g(Y)] = \mathbb{E}[f(X)]\mathbb{E}[g(Y)]$$

Proof. Let the random variable $Z = (X, Y)$, so $h(Z) = f(X)g(Y)$. Then

$$\begin{aligned}
\mathbb{E}[h(Z)] &= \sum_{(x,y)} h(x,y)\mathbb{P}(Z = (x,y)) \\
&= \sum_{(x,y)} f(x)g(y)\mathbb{P}(X = x, Y = y) \\
&= \sum_{(x,y)} f(x)g(y)\mathbb{P}(X = x)\mathbb{P}(Y = y) \\
&= \mathbb{E}[f(X)] \cdot \mathbb{E}[g(Y)]
\end{aligned}$$

■

Proposition 4.27

If X and Y are independent, $\mathrm{Cov}(X, Y) = 0$.

> **Remark.** The converse is not true.

Example 4.28

Let X_1, X_2, X_3 be independent Bernoulli random variables with $p = \frac{1}{2}$. Let $Y_1 = 2X_1 - 1$, $Y_2 = 2X_2 - 1$. Also let $Z_1 = Y_1X_3$, $Z_2 = Y_2X_3$. Then $\mathbb{E}[Y_1] = \mathbb{E}[Y_2] = 0$, and $\mathbb{E}[Z_1] = \mathbb{E}[Z_2] = 0$ by independence. Then

$$\mathrm{Cov}(Z_1, Z_2) = \mathbb{E}[Z_1Z_2] = \mathbb{E}[Y_1Y_2X_3^2]$$

But we claim Z_1 and Z_2 are not independent. Observe $\mathbb{P}(Z_1 = 0, Z_2 = 0) = \mathbb{P}(X_3 = 0) = \frac{1}{2}$. But $\mathbb{P}(Z_1 = 0) = \mathbb{P}(Z_2 = 0) = \frac{1}{2}$.

§4.5 Inequalities on random variables

Proposition 4.29 (Markov's inequality)
Let X be a non-negative random variable and $a > 0$. Then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof. We have

$$\begin{aligned} & X \geq a \cdot 1(X \geq a) \\ \implies & \mathbb{E}[X] \geq a \cdot \mathbb{P}(X \geq a) \end{aligned}$$

by taking expectations. ■

Proposition 4.30 (Chebyshev's inequality)
Let X be a random variable with $\mathbb{E}[X] < \infty$. Then, for any $a > 0$,

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Proof. Observe

$$\begin{aligned} \mathbb{P}(|X - \mathbb{E}[X]| \geq a) &= \mathbb{P}((X - \mathbb{E}[X])^2 \geq a^2) \\ &\leq \frac{\mathbb{E}[(X - \mathbb{E}[X])^2]}{a^2} && \text{by Markov} \\ &= \frac{\mathrm{Var}(X)}{a^2} \end{aligned}$$

■

Proposition 4.31 (Cauchy-Schwarz)
Let X and Y be random variables, then

$$\mathbb{E}[|XY|] \leq \sqrt{\mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]}$$

Proof. Assume that $\mathbb{E}[X^2]$ and $\mathbb{E}[Y^2]$ are finite, otherwise there is nothing to prove. Then, $|XY| \leq \frac{1}{2}(X^2 + Y^2)$, so, taking expectations, $\mathbb{E}[|XY|]$ is also finite. We may also assume $\mathbb{E}[X^2] > 0$, $\mathbb{E}[Y^2] > 0$. Let us also assume for the time being that $X, Y \geq 0$.

Let $t \in \mathbb{R}$ and consider $(X - tY)^2$. We have

$$\begin{aligned} 0 &\leq \mathbb{E}[(X - tY)^2] \\ &= \mathbb{E}[X^2] - 2t\mathbb{E}[XY] + t^2\mathbb{E}[Y^2] := f(t) \end{aligned}$$

Then $f'(t) = -2\mathbb{E}[XY] + 2t\mathbb{E}[Y^2]$, so $\min f(t)$ is achieved at $t^* = \frac{\mathbb{E}[XY]}{\mathbb{E}[Y^2]}$. We hence have

$$0 \leq \mathbb{E}[X^2] - 2\frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]} + \frac{\mathbb{E}[XY]^2}{\mathbb{E}[Y^2]}$$

$$\implies \mathbb{E}[XY]^2 \leq \mathbb{E}[X^2] \cdot \mathbb{E}[Y^2]$$

■

Remark. We could also have just put in our value for t^* from the beginning – the minimisation approach is purely to provide rationale.

Remark. When is equality achieved? We would need $f(t^*) = 0$, that is

$$\begin{aligned} &\mathbb{E}[(X - t^*Y)^2] = 0 \\ \implies &P(X = t^*Y) = \mathbb{P}(X - t^*Y = 0) = 1 \end{aligned}$$

so X is just a multiple of Y.

Definition 4.32 (Convexity)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is called *convex* if, for any $x, y \in \mathbb{R}$ and any $t \in [0, 1]$,

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y)$$

Proposition 4.33 (Jensen's inequality)

Let X be a random variable, and $X : \mathbb{R} \to \mathbb{R}$ a convex function. Then

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X])$$

Remark. We can remember the direction of the inequality by putting $f(x) = x^2$ and using non-negativity of the variance.

Lemma 4.34

Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function. Then f is the supremum of all the lines lying below it. In other words, for all m in $\mathbb{R}$, there exist a, b in $\mathbb{R}$ such that $f(m) = am+b$ and $f(x) \geq ax + b$ for all x.

Proof of lemma. Let $m \in \mathbb{R}$, and suppose $x < m < y$. Then $m = tx + (1 - t)y$ for some $t \in [0, 1]$, or, equivalently,

$$t(m - x) = (1 - t)(y - m) \qquad (*)$$

By convexity, we have

$$\begin{aligned} f(m) = f(tx + (1 - t)y) &\leq tf(x) + (1 - t)f(y) \\ \implies t(f(m) - f(x)) &\leq (1 - t)(f(y) - f(m)) \\ (*) \implies \frac{f(m) - f(x)}{m - x} &\leq \frac{f(y) - f(m)}{y - m} \end{aligned}$$

Then set

$$a = \sup_{x<m} \frac{f(m) - f(x)}{m - x}$$

so that, for any $x < m < y$,

$$\frac{f(m) - f(x)}{m - x} \le a \le \frac{f(y) - f(m)}{y - m}$$

Hence, by rearrangement, we conclude $f(z) \ge a(z - m) + f(m)$, for any z (depending on whether z is greater than m or less than m, we use either the left or right inequality.) ■

Proof of proposition. Let $m = \mathbb{E}[X]$. By the claim, we have $a, b \in \mathbb{R}$ such that $f(m) = am + b$ and $f(x) \ge ax + b$ for all x. We have

$$\begin{aligned} & f(X) \ge aX + b \\ \implies & \mathbb{E}[f(X)] \ge a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X]) \end{aligned}$$

■

Remark. When is equality achieved?

Suppose f convex with the extra property that, for $n = \mathbb{E}[X]$, there exist $a, b \in \mathbb{R}$ with $f(m) = am + b$ but $f(x) > ax + b$, note the strict inequality, for $x \ne m$. When should X satisfy $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$?

Since $f(X) \ge aX + b$, we have $f(X) - (aX + b) \ge 0$. Taking expectations, we have

$$\mathbb{E}[f(X)] \ge a\mathbb{E}[X] + b = am + b = f(m) = f(\mathbb{E}[X])$$

Since we want $\mathbb{E}[f(X)] = f(\mathbb{E}[X])$, we must have

$$\begin{aligned} & \mathbb{E}[f(X) - (aX + b)] = 0 \\ \implies & \mathbb{P}(f(X) = aX + b) = 1 \end{aligned}$$

but then, since $f(x) > ax + b$ for $x \ne m$, we conclude $\mathbb{P}(X = m) = 1$ (if we didn't have that extra part, we'd conclude f is just a linear function).

Lecture $n + 1$

Proposition 4.35 (AM-GM inequality)

Let $x_1, \dots, x_n > 0$. Then

$$\left(\prod_{k=1}^{n} x_k\right)^{\frac{1}{n}} \le \frac{1}{n}\sum_{k=1}^{n} x_k$$

Lemma 4.36

Let f be a convex function. Then, for any $x_1, \dots, x_n \in \mathbb{R}$ we have

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) \ge f\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right)$$

Proof of lemma. Let X be a random variable with $\mathbb{P}(X = x_i) = \frac{1}{n}$, $i = 1, \dots, n$. Then

$$\frac{1}{n}\sum_{k=1}^{n} f(x_k) = \sum_x f(x)\mathbb{P}(X = x) = \mathbb{E}[f(X)]$$

so Jensen's gives

$$\mathbb{E}[f(X)] \geq f(\mathbb{E}[X]) = f\left(\frac{1}{n}\sum_{k=1}^{n} x_k\right)$$

as required. ■

Proof of proposition. Let $f(x) = -\log x$ and suppose $x_1, \dots, x_n > 0$. Then

$$-\frac{1}{n}\sum_{k=1}^{n} \log x_k \geq -\log\left(\frac{1}{n}\sum_{k=1}^{n} x_n\right)$$
$$\implies \left(\prod_{k=1}^{n} x_k\right)^{\frac{1}{n}} \leq \frac{1}{n}\sum_{k=1}^{n} x_k$$

as required. ■

§4.6 Joint distribution and conditional expectation

Recall if X is a discrete random variable, then the disttribution of X is $(\mathbb{P}(X = x))_x$.

Definition 4.37. Let $X_1, \dots, X_n$ be discrete random variables. Their *joint distribution* is defined to be the set of values of

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$$

for any $x_1, \dots, x_n$.

> **Remark.** We can recover the distribution of any one of the variables from such a joint distribution with
>
> $$\mathbb{P}(X_1 = x_1) = \sum_{x_2, \dots, x_n} \mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$
>
> In this context the distribution of the single variable is sometimes called its *marginal* distribution.

Definition 4.38. Let X and Y be discrete random variables. The *conditional distribution* of X given $Y = y$ is

$$\mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}$$

By the law of probability we of course have

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x|Y = y)\mathbb{P}(Y = y)$$

Let X and Y be independent – what is the distribution of $X+Y$? We have

$$\begin{aligned}\mathbb{P}(X+Y=z) &= \sum_y \mathbb{P}(X+Y=z, Y=y) \\ &= \sum_y \mathbb{P}(X=z-y, Y=y) \\ &= \sum_y \mathbb{P}(X=z-y)\mathbb{P}(Y=y)\end{aligned}$$

where the last line is by independence. This is of course a convolution.

Example

Let $X \sim \text{Poi}(\lambda)$ and $Y \sim \text{Poi}(\mu)$, with $\lambda, \mu > 0$ and $X \perp Y$. Then

$$\begin{aligned}\mathbb{P}(X+Y=n) &= \sum_{k=0}^{n} \mathbb{P}(X=n-k)\mathbb{P}(Y=k) \\ &= e^{-\lambda}\frac{\lambda^{n-k}}{(n-k)!} \cdot e^{-\mu}\frac{\mu^k}{k!} \\ &= \frac{e^{-(\lambda+\mu)}}{n!}\sum_{k=0}^{n}\binom{n}{k}\lambda^{n-k}\mu^k \\ &= e^{-(\lambda+\mu)}\frac{(\lambda+\mu)^n}{n!}\end{aligned}$$

so in particular $X+Y \sim \text{Poi}(\lambda+\mu)$.

Let us derive conditional expectation for random variables. First, take $B \in \mathcal{F}$ to be an event with $\mathbb{P}(B) > 0$. We have

$$\mathbb{E}[X|B] = \frac{\mathbb{E}[X \cdot 1(B)]}{\mathbb{P}(B)}$$

and, if $X = 1(A)$, then

$$\mathbb{E}[1(A)|B] = \frac{\mathbb{E}[1(A)\cdot 1(B)]}{\mathbb{P}(B)} = \frac{\mathbb{E}[1(A\cap B)]}{\mathbb{P}(B)} = \mathbb{P}(A|B)$$

We can now formulate the *law of total expectation.*

Proposition 4.39 (Law of total expectation)

Let X be a random variable and (Ω_n) a partition of Ω with $\mathbb{P}(\Omega_n) > 0$ for any n. Then

$$\mathbb{E}[X] = \sum_n \mathbb{E}[X|\Omega_n]\mathbb{P}(\Omega_n)$$

Proof. Use indicators. We have

$$\mathbb{E}[X] = \mathbb{E}[X \cdot 1(\Omega)] = \mathbb{E}\left[X \cdot \sum_n 1(\Omega_n)\right] = \sum_n \mathbb{E}[X|\Omega_n]\mathbb{P}(\Omega_n)$$

■

Now, we want to condition expectation not on an event but on a particular random variable. We have

$$\begin{aligned}\mathbb{E}[X|Y=y] &= \frac{\mathbb{E}[X \cdot 1(Y=y)]}{\mathbb{P}(Y=y)} \\ &= \sum_x x \cdot \frac{\mathbb{P}(X=x, Y=y)}{\mathbb{P}(Y=y)} \\ &= \sum_x x \cdot \mathbb{P}(X=x|Y=y)\end{aligned}$$

Let us now define $g(y) = \mathbb{E}[X|Y=y]$.

Definition 4.40. The expectation of X given Y, $\mathbb{E}[X|Y]$, is a random variable which is a function of Y. In particular,

$$\mathbb{E}[X|Y] = g(Y)$$

Example

Toss a p-coin independently n times, and let $X_i = 1(i^{\text{th}} \text{ toss is H})$. Let $Y_n = X_1 + \cdots + X_n$. Let us attempt to calculate $\mathbb{E}[X_1|Y_n]$. First, let

$$\begin{aligned}g(y) = \mathbb{E}[X_1|Y_n = y] &= \frac{\mathbb{E}[X_1 \cdot 1(Y_n = y)]}{\mathbb{P}(Y_n = y)} \\ &= \frac{\mathbb{P}(X_1 = 1, Y_n = y)}{\mathbb{P}(Y_n = y)} \\ &= \frac{p\binom{n-1}{y-1}p^{y-1}(1-p)^{n-y}}{\binom{n}{y}p^y(1-p)^{n-y}} \\ &= \frac{y}{n}\end{aligned}$$

Hence, $\mathbb{E}[X_1|Y_n] = \frac{Y_n}{n}$.

We have some simple and uninspiring properties:

- $\mathbb{E}[cX|Y] = c\mathbb{E}[X|Y]$;
- $\mathbb{E}[c + X|Y] = c + \mathbb{E}[X|Y]$;
- $\mathbb{E}[c|Y] = c$;
- If $X_1, \ldots, X_n$ are random variables,

$$\mathbb{E}\left[\sum_{i=1}^n X_i \middle| Y\right] = \sum_{i=1}^n \mathbb{E}[X_i|Y]$$

and some more interesting properties. We have

$$\begin{aligned}\mathbb{E}[X|Y] &= \sum_y 1(Y=y)\mathbb{E}[X|Y=y] \\ \implies \mathbb{E}[\mathbb{E}[X|Y]] &= \sum_y \mathbb{E}[X|Y=y]\mathbb{P}(Y=y) = \mathbb{E}[X]\end{aligned}$$

Suppose now X and Y are independent random variables. Then

$$\begin{aligned}\mathbb{E}[X|Y] &= \sum_y 1(Y=y)\mathbb{E}[X|Y=y] \\ &= \sum_y 1(Y=y)\sum_x x\cdot\mathbb{P}(X=x|Y=y) \\ &= \sum_y 1(Y=y)\sum_x x\cdot\mathbb{P}(X=x) \\ &= \sum_y 1(Y=y)\mathbb{E}[X] = \mathbb{E}[X]\end{aligned}$$

Proposition 4.41

Let Y and Z be independent. Then

$$\mathbb{E}[\mathbb{E}[X|Y]|Z] = \mathbb{E}[X]$$

Proof. Let $g(Y) = \mathbb{E}[X|Y]$, $g(y) = \mathbb{E}[X|Y=y]$. We claim $g(Y)$ is independent of Z. Observe

$$\begin{aligned}\mathbb{P}(g(Y)=w, Z=z) &= \sum_{y\in g^{-1}(\{w\})} \mathbb{P}(Y=y, Z=z) \\ &= \sum_{y\in g^{-1}(\{w\})} \mathbb{P}(Y=y)\mathbb{P}(Z=z) \\ &= \mathbb{P}(g(Y)=w)\mathbb{P}(Z=z)\end{aligned}$$

Hence, previous results give $\mathbb{E}[g(Y)|Z] = \mathbb{E}[g(Y)]$, and this is $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$. ∎

Proposition 4.42

Let X, Y be two random variables, and $h : \mathbb{R} \to \mathbb{R}$. Then

$$\mathbb{E}[h(Y)\cdot X|Y] = h(Y)\cdot\mathbb{E}[X|Y]$$

Proof. We have

$$\begin{aligned}\mathbb{E}[h(Y)\cdot X|Y] &= \sum_y 1(Y=y)\mathbb{E}[h(Y)\cdot X|Y=y] \\ &= \sum_y 1(Y=y)h(y)\mathbb{E}[X|Y=y] \\ &= h(Y)\sum_y 1(Y=y)\mathbb{E}[X|Y=y] \\ &= h(Y)\mathbb{E}[X|Y]\end{aligned}$$

∎

Corollary 4.43

$$\mathbb{E}[\mathbb{E}[X|Y]|Y] = \mathbb{E}[X|Y]$$

Example

Toss a p-coin independently n times, and let $X_i = 1(i^{\text{th}}$ toss is H). Let $Y_n = X_1 + \cdots + X_n$. What is $\mathbb{E}[X_1|Y_n]$?

By symmetry, $\mathbb{E}[X_1|Y_n] = \mathbb{E}[X_i|Y_n]$ for any i, $1 \leq i \leq n$. Then

$$Y_n = \mathbb{E}[Y_n|Y_n] = \mathbb{E}[X_1 + \cdots + X_n|Y_n] = n\mathbb{E}[X_1|Y_n]$$

which was what we wanted.

§5 Random walks

Definition 5.1 (Random process)**.** A *random process* or *stochastic process* is a sequence of random variables (X_n).

Definition 5.2 (Random walks)**.** A *random walk* is a random process that can be expressed as

$$X_n = x + Y_1 + \cdots + Y_n$$

where x is a number and (Y_i) are independent identically distributed (i.i.d.) random variables. We call the Y_i the *steps* of the random variable.

A *simple random walk* has

$$Y_i = \begin{cases} 1 & \text{with probability } p \\ -1 & \text{with probability } q = 1 - p \end{cases}$$

We will write

- $\mathbb{P}_x(A) = \mathbb{P}(A|X_0 = x)$;
- $h(x) = \mathbb{P}_x((X_n)$ reaches a before 0), where a is some arbitrary constant at which the game ends.

so $h(0) = 0$, $h(a) = 1$. Then

$$\begin{aligned} h(x) &= \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0) \\ &= \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0|Y_1 = 1) \cdot p + \mathbb{P}_x((X_n) \text{ hits } a \text{ bef. } 0|Y_1 = -1) \cdot q \\ &= p \cdot h(x+1) + q \cdot h(x-1) \end{aligned}$$

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- *Case* $p = q = \frac{1}{2}$, then X_n is a *simple symmetric random walk*. We have

$$\begin{aligned} h(x) &= \frac{1}{2}h(x+1) + \frac{1}{2}h(x-1) \\ \implies h(x+1) - h(x) &= h(x) - h(x-1) \\ \implies &\boxed{h(x) = cx} \end{aligned}$$

  for some constant c. Since $h(a) = 1$, $c = \frac{1}{a}$, and hence $h(x) = \frac{x}{a}$.

- *Case* $p \neq q$. Let us try a solution of the form λ^x for some lambda – we have

$$\begin{aligned} \lambda^x &= p\lambda^{x+1} + q\lambda^{x-1} \\ \implies \lambda &= 1, \lambda = \frac{q}{p} \end{aligned}$$

  so the general solution has the form

$$h(x) = A + B\left(\frac{q}{p}\right)^x$$

  which, applying boundary conditions, gives

$$\boxed{h(x) = \frac{\left(\frac{q}{p}\right)^x - 1}{\left(\frac{q}{p}\right)^a - 1}}$$

These results are called the *gambler's ruin estimates*.

Now let us consider a new quantity – let $T = \min\{n \geq 0 : X_n \in \{0, a\}\}$ – the first n such that X_n is either 0 or a, i.e. the duration of the game. We want to find $\mathbb{E}[T|X_0 = x] = \mathbb{E}_x[T] = k(x)$. We have from the law of total expectation

$$\begin{aligned} k(x) &= \mathbb{E}_x[T|Y_1 = 1] \cdot \mathbb{P}(Y_1 = 1) + \mathbb{E}_x[T|Y_1 = -1] \cdot \mathbb{P}(Y_1 = -1) \\ &= p(1 + k(x+1)) + q(1 + k(x-1)) \\ &= 1 + pk(x+1) + qk(x-1) \end{aligned}$$

with $k(0) = k(a) = 0$.

- *Case* $p = q = \frac{1}{2}$. Try a solution of the form Ax^2, to get

$$\begin{aligned} Ax^2 &= 1 + pA(x+1)^2 + qA(x-1)^2 \\ \implies A &= -1 \end{aligned}$$

  giving a general solution of the form $k(x) = -x^2 + Bx + C$. Applying boundary conditions, we obtain

$$\boxed{k(x) = x(a - x)}$$

- *Case* $p \neq q$. Try $k(x) = Cx$ to get

$$\begin{aligned} Cx &= 1 + pC(x+1) + qC(x-1) \\ \implies c &= \frac{1}{q-p} \end{aligned}$$

Adding this to the solution of the homogeneous equation, we obtain

$$k(x) = A + \frac{1}{q-p}x + B\left(\frac{q}{p}\right)^x$$

which, after boundary conditions, gives

$$\boxed{k(x) = \frac{1}{q-p}x - \frac{a}{q-p} \cdot \frac{\left(\frac{q}{p}\right)^x - 1}{\left(\frac{q}{p}\right)^a - 1}}$$

§6 Probability generating functions

Let X be a random variable with values in $\mathbb{N}$ – recall $p_r = \mathbb{P}(X = r)$ for $r \in \mathbb{N}$ constitutes the probability mass function.

Definition 6.1 (Probability generating function)**.** The *probability generating function (PGF)* for a random variable X with values in $\mathbb{N}$ and probability masses (p_r) is

$$p(z) = \mathbb{E}[z^X] = \sum_{r=0}^{\infty} p_r z^r$$

with $|z| \leq 1$.

Remark. This is well-defined because

$$|p(z)| = \left|\sum_{r=0}^{\infty} p_r z^r\right| \leq \sum_{r=0}^{\infty} p_r |z|^r \leq \sum_{r=0}^{\infty} p_r = 1$$

so p converges absolutely for $|z| \leq 1$.

Theorem 6.2

The pgf uniquely determines the distribution of X.

Proof. Let (p_r) and (q_r) be two probability distributions with the same pgf's – for all $|z| \leq 1$, $\sum p_r z^r = \sum q_r z^r$.

Plugging in $z = 0$ gives $p_0 = q_0$. We now proceed by strong induction. Suppose $p_r = q_r$ for all $r \leq n$, so

$$\sum_{r=n+1}^{\infty} p_r z^r = \sum_{r=n+1}^{\infty} q_r z^r$$

Clearly dividing by z^{n+1} and setting $z = 0$ gives $p_{n+1} = q_{n+1}$ as required. ∎

Proposition 6.3

Suppose $X_1, \ldots, X_n$ are independent random variables with pgf's $q_i(z)$, and let $X = X_1 + \cdots + X_n$ have pgf p. Then

$$p(z) = q_1(z) \ldots q_n(z)$$

Proof. Clearly

$$p(z) = \mathbb{E}[z^{X_1+\dots+X_n}] = \mathbb{E}[X_1]\dots\mathbb{E}[X_n] = q_1(z)\dots q_n(z)$$

by independence. ∎

Example

Let us calculate some pgf's for important distributions.

① *Binomial*. Suppose $X \sim \mathrm{Bin}(n,p)$. Then

$$\begin{aligned} p(z) &= \sum_{k=0}^{n} z^k \binom{n}{k} p^k (1-p)^{n-k} \\ &= (pz+1-p)^n \end{aligned}$$

② *Geometric*. Suppose $X \sim \mathrm{Geo}(p)$. Then

$$\begin{aligned} \mathbb{E}[z^X] &= \sum_{r=0}^{\infty} z^r (1-p)^r p \\ &= \frac{p}{1-z(1-p)} \end{aligned}$$

③ *Poisson*. Suppose $X \sim \mathrm{Poi}(\lambda)$. Then

$$\begin{aligned} \mathbb{E}[z^X] &= \sum_{r=0}^{\infty} z^r e^{-\lambda} \frac{\lambda^r}{r!} \\ &= e^{\lambda(z-1)} \end{aligned}$$

Example

This fact can make calculating the pmf of sums of random variables easier. For example, suppose $X \sim \mathrm{Bin}(n,p)$ and $Y \sim \mathrm{Bin}(m,p)$ with $X \perp Y$. Then

$$\begin{aligned} \mathbb{E}[z^{X+Y}] = \mathbb{E}[z^X]\mathbb{E}[z^Y] &= (pz+1-p)^n \cdot (pz+1-p)^m \\ &= (pz+1-p)^{n+m} \end{aligned}$$

from which it follows that $X+Y \sim \mathrm{Bin}(n+m,p)$.

What about Poisson? Suppose $X \sim \mathrm{Poi}(\lambda)$, $Y \sim \mathrm{Poi}(\mu)$, and $X \perp Y$. Then

$$\mathbb{E}[z^{X+Y}] = e^{\lambda(z-1)} e^{\mu(z-1)} = e^{(\lambda+\mu)(z-1)}$$

from which it follows that $X+Y \sim \mathrm{Poi}(\lambda+\mu)$.

For the following results, it will be helpful to introduce the notation

$$p(1^-) = \lim_{z \to 1^-} p(z)$$

Proposition 6.4

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{E}[X] = p'(1^-)$$

Proof.

- Assume $\mathbb{E}[X] < \infty$, and take $0 < z < 1$. We have

$$p'(z) = \sum_{r=1}^{\infty} r p_r z^{r-1} \leq \sum_{r=1}^{\infty} r p_r = \mathbb{E}[X]$$

Hence p' is upper bounded by $\mathbb{E}[X]$, and it is clearly increasing, so the limit exists and is $\leq \mathbb{E}[X]$.

Let $\varepsilon > 0$. There exists N such that

$$\sum_{r=1}^{N} r p_r \geq \mathbb{E}[X] - \varepsilon$$

since this is what it means for the infinite sum to be equal to $\mathbb{E}[X]$. Furthermore,

$$\lim_{z \to 1^-} p'(z) \geq \lim_{z \to 1^-} \sum_{r=1}^{N} r p_r z^{r-1} = \sum_{r=1}^{N} r p_r \geq \mathbb{E}[X] - \varepsilon$$

- If $\mathbb{E}[X] = \infty$, then, for any $M \geq 0$, we have N such that

$$\sum_{r=1}^{N} r p_r \geq M$$

Then, as above,

$$\lim_{z \to 1^-} p'(z) \geq \lim_{z \to 1^-} \sum_{r=1}^{N} r p_r z^{r-1} = \sum_{r=1}^{N} r p_r \geq M$$

so the limit is infinite.

∎

Proposition 6.5

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{E}[X(X-1)\dots(X-k+1)] = p^{(k)}(1^-)$$

Corollary 6.6

Let X be a discrete random variable with pgf $p(z)$, then

$$\text{Var}(X) = p''(1^-) + p'(1^-) - (p'(1^-))^2$$

Proposition 6.7

Let X be a discrete random variable with pgf $p(z)$, then

$$\mathbb{P}(X = n) = \frac{1}{n!} \frac{d^n p}{dz^n}\bigg|_{z=0}$$

Remark. This is another way to see theorem 6.2.

§6.1 Sum of a random number of random variables

Let $X_1, X_2, \ldots$ be i.i.d. random variables and let N be an independent random variable with values in $\mathbb{N}$. Set

$$S_n = X_1 + \cdots + X_n$$

that, is

$$S_n(\omega) = \sum_{i=1}^{n} X_i(\omega)$$

Passing to the random variable N, we obtain

$$S_N(\omega) = S_{N(\omega)}(\omega) = \sum_{i=1}^{N(\omega)} X_i(\omega)$$

Lemma 6.8

Let $q(z) = \mathbb{E}[z^N]$ be the pgf of N, and $p(z) = \mathbb{E}[z^{X_i}]$ be the pgf of X_i. Then

$$r(z) = \mathbb{E}[z^{S_N}] = q(p(z))$$

Proof 1. We have

$$\begin{aligned}
r(z) = \mathbb{E}[z^{S_N}] &= \sum_{n=0}^{\infty} \mathbb{E}[z^{S_N} \cdot 1(N = n)] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[z^{S_n} \cdot 1(N = n)] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[z^{X_1 + \cdots + X_n} \cdot 1(N = n)] \\
&= \sum_{n=0}^{\infty} \mathbb{E}[z^{X_1 + \cdots + X_n}]\mathbb{P}(N = n) \\
&= \sum_{n=0}^{\infty} (p(z))^n \mathbb{P}(N = n) \\
&= q(p(z))
\end{aligned}$$

■

Proof 2. We have

$$r(z) = \mathbb{E}[z^{S_n}] = \mathbb{E}[\mathbb{E}[z^{S_N}|N]] = g(N)$$

where

$$\begin{aligned} g(n) &= \mathbb{E}[z^{S_N}|N = n] \\ &= \mathbb{E}[z^{S_n}] \\ &= p(z)^n \end{aligned}$$

so that

$$\begin{aligned} r(z) &= \mathbb{E}[p(z)^n] \\ &= q(p(z)) \end{aligned}$$

■

Corollary 6.9
We have

$$\begin{aligned} \mathbb{E}[S_n] &= \mathbb{E}[N] \cdot \mathbb{E}[X_1] \\ \mathrm{Var}(S_n) &= \mathbb{E}[N] \cdot \mathrm{Var}(X_1) + \mathrm{Var}(N) \cdot \mathbb{E}[X_1]^2 \end{aligned}$$

Proof. The first result follows from

$$\begin{aligned} \mathbb{E}[S_n] &= r'(1-) \\ &= \lim_{z \to 1^-} r'(z) \\ &= \lim_{z \to 1^-} q'(p(z)) \lim_{z \to 1^-} p'(z) \end{aligned}$$

Clearly $p(z)$ approaches 1 from below as z approaches 1 from below, so this is just

$$q'(1-)p'(1-) = \mathbb{E}[N] \cdot \mathbb{E}[X_1]$$

as required.

The second result is left as an exercise. ■

§7 Branching processes

Think of a branching process as a series of generations of offspring. We have $X_0 = 1$, there being one person in the first generation; then, in each generation, each individual produces k offspring independently with a fixed distribution (in particular, the distribution of X_1).

Let $Y_{k,n}$ with $k \geq 1$, $n \geq 0$ be i.i.d. random variables with the distribution of X_1, interpreted as the number of offspring produced by the k^{th} individual in the n^{th} generation. Hence

$$X_{n+1} = \begin{cases} Y_{1,n} + \cdots + Y_{X_n,n} & X_n \geq 1 \\ 0 & X_n = 0 \end{cases}$$

Proposition 7.1

We have

$$\mathbb{E}[X_n] = \mathbb{E}[X_1]^n$$

Proof. We will proceed by induction. The base case is clear. Let us consider the conditional expectation $\mathbb{E}[X_{n+1}|X_n = m]$ – we have

$$\begin{aligned}
\mathbb{E}[X_{n+1}|X_n = m] &= \mathbb{E}[Y_{1,n} + \cdots + Y_{m,n}|X_n = m] \\
&= \mathbb{E}[Y_{1,n} + \cdots + Y_{m,n}] && \text{(independence)} \\
&= m\mathbb{E}[X_1]
\end{aligned}$$

so that $\mathbb{E}[X_{n+1}|X_n] = X_n\mathbb{E}[X_1]$. Hence $\mathbb{E}[X_{n+1}] = \mathbb{E}[\mathbb{E}[X_{n+1}|X_n]] = \mathbb{E}[X_n\mathbb{E}[X_1]] = \mathbb{E}[X_n]\mathbb{E}[X_1]$, which completes the result by induction. ■

Proposition 7.2

Let $G(z) = \mathbb{E}[z^{X_1}]$, $G_n(z) = \mathbb{E}[z^{X_n}]$, then

$$G_{n+1}(z) = G(G_n(z))$$

Proof. Again, let us consider $\mathbb{E}[z^{X_{n+1}}|X_n = m]$. We have

$$\begin{aligned}
\mathbb{E}[z^{X_{n+1}}|X_n = m] &= \mathbb{E}[z^{Y_{1,n}+\cdots+Y_{m,n}}|X_n = m] \\
&= \mathbb{E}[z^{Y_{1,n}+\cdots+Y_{m,n}}] \\
&= (G(z))^m
\end{aligned}$$

Hence $G_{n+1}(z) = \mathbb{E}[\mathbb{E}[z^{X_{n+1}}|X_n]] = \mathbb{E}[G(z)^{X_n}] = G_n(G(z))$ ■

§7.1 Extinction probability

Let $q_n = \mathbb{P}(X_n = 0)$, and $q = \mathbb{P}(X_n = 0 \text{ for some } n \geq 0)$. Letting A_n be the event that $X_n = 0$ for some n, we can see $q = \mathbb{P}(\cup A_n)$, and $A_n \subseteq A_{n+1}$, so the A_n form an increasing sequence and we get

$$\mathbb{P}(A_n) \to \mathbb{P}\left(\bigcup A_n\right)$$

by proposition 1.8 (continuity of probability measures).

Lemma 7.3

Let $G(z) = \mathbb{E}[z^{X_1}]$, then $q_{n+1} = G(q_n)$ and $q = G(q)$.

Proof 1. We have

$$\begin{aligned}
q_{n+1} &= \mathbb{P}(X_{n+1} = 0) \\
&= G_{n+1}(0)
\end{aligned}$$

$$= G(G_n(0))$$
$$= G(q_n)$$

Then, since G is continuous and $q_n \to q$, we can simply take the limit on both sides to get $q = G(q)$ as required. ■

Proof 2. We will condition q_{n+1} on the number of individuals in the first generation, i.e. X_1. Supposing $X_1 = m$, let $X_n^{(1)}, \dots, X_n^{(m)}$ be i.i.d. branching processes with the same distribution as the original one. Hence

$$X_{n+1} = X_n^{(1)} + \cdots + X_n^{(m)}$$

So

$$\begin{aligned}
q_{n+1} &= \mathbb{P}(X_{n+1} = 0) \\
&= \sum_m \mathbb{P}(X_{n+1} = 0 | X_1 = m)\mathbb{P}(X_1 = m) \\
&= \sum_m \mathbb{P}(X_n^{(1)} + \cdots + X_n^{(m)} = 0 | X_1 = m)\mathbb{P}(X_1 = m) \\
&= \sum_m \left(\mathbb{P}(X_n^{(1)} = 0)\right)^m \mathbb{P}(X_1 = m) \\
&= \sum_m q_n^m \mathbb{P}(X_1 = m) \\
&= G(q_n)
\end{aligned}$$

Once again, we take the limit on both sides. ■

Lecture 14 – 23/02/26

Recalling proposition 7.1, we have that

- if $\mathbb{E}[X_1] < 1$, $\mathbb{E}[X_n] \to 0$ as $n \to \infty$, so we expect $q = 1$;
- if $\mathbb{E}[X_1] > 1$, $\mathbb{E}[X_n] \to \infty$ as $n \to \infty$, so we expect $q < 1$;
- if $\mathbb{E}[X_1] = 1$, then $\mathbb{E}[X_n] = 1$ for all n. But it is not clear what q will be. Let us focus on this last case.

We can also interpret these results in the context of a graph of G. We know that the graph of G will intersect x at $x = 1$, and we could either have another intersection in the range $0 \leq x < 1$, or no intersections.

- *There is another intersection.* Then iterating G starting from $G(0)$ will lead us to the other intersection. Furthermore, the gradient of G at 1, $G(1-) = \mathbb{E}[X_1]$ must be greater than 1 for there to be such an intersection, as we expect;
- *There is not another intersection.* Then iterating G will take us all the way to 1, and the gradient of G at 1, $G(1-) = \mathbb{E}[X_1]$ must be less than or equal to 1 for there not to be any further intersections. We conclude that $q = 1$ in the case $\mathbb{E}[X_1] = 1$.

Proposition 7.4

Assume $\mathbb{P}(X_1 = 1) < 1$. The extinction probability q is the minimal non-negative solution to $G(x) = x$. Furthermore, $q < 1$ if and only if $\mathbb{E}[X_1] > 1$.

Proof. We know $q = G(q)$. Let t be the smallest minimal solution of $x = G(x)$. We will show by induction that, for any n, $q_n \leq t$, and then take limits to establish $q \leq t \implies q = t$.

For the base case, $0 = q_0 \leq t$. Suppose $q_n \leq t$, then

$$q_{n+1} = G(q_n) \leq G(t) = t$$

since G is increasing.

Now, we show $q < 1$ if and only if $\mathbb{E}[X_1] > 1$. Suppose for now $g_0 + g_1 = \mathbb{P}(X_1 \leq 1) < 1$.

Let $H(z) = G(z) - z = \sum_{r=0}^{\infty} g_r z^r - z$, so $H(1) = 0$. First, we show that H can have at most one root in $[0, 1)$. Observe

$$H''(z) = \sum_{r=0}^{\infty} r(r-1)g_r z^{r-2} > 0$$

in $(0, 1)$, because $g_0 + g_1 < 1$, so there must be some later r with a positive value of g_r, making the sum strictly larger than zero. Hence $H'(z)$ is strictly increasing, and so can have at most one more root other than 1 – otherwise, we would have more than one point c with $H'(c) = 0$ by Rolle's theorem.

We now split into two cases.

- H has no root other than 1, so $q = 1$. Then $H'(1-) = G'(1-) - 1 = \mathbb{E}[X_1] - 1$. Furthermore, $H(0) = G(0) = g_0 \geq 0$ while $H(1) = 0$, so $H(z) \geq 0$ for all $z \in [0, 1]$. Now

  $$H'(1-) = \lim_{z \to 1^-} \frac{H(z) - H(1)}{z - 1} \leq 0$$

  since the denominator is $H(z) - 0 \geq 0$ while the denominator is $z - 1 \leq 0$. Hence $H'(1-) \leq 0 \implies \mathbb{E}[X_1] \leq 1$, as required.

- H has one more root r, so r is the extinction probability. We hence have $H(r) = 0$ and $H(1) = 0$ with $r < 1$, so we have $z \in (r, 1)$ with $H'(z) = 0$ by Rolle. Hence $\mathbb{E}[X_1] - 1 = H'(-1) > H'(z) = 0$, so $\mathbb{E}[X_1] > 1$ as required.

On the other hand, if $\mathbb{P}(X_1 \leq 1) = 1$, then $\mathbb{E}[X_1] = g_1$, so

$$\begin{aligned} G(z) &= g_0 + g_1 z \\ &= 1 - \mathbb{E}[X_1] + \mathbb{E}[X_1]z \end{aligned}$$

Then, if $G(z) = z$, we have $(1 - \mathbb{E}[X_1])z = 1 - \mathbb{E}[X_1]$ and, since $\mathbb{E}[X_1] = g_1 < 1$, $1 - \mathbb{E}[X_1] \neq 0$ so $z = 1$.

■

§8 Continuous random variables

Recall definition 4.2, that

Definition. Given a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, a *random variable* is a function

$$X : \Omega \to \mathbb{R}$$

satisfying

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F}$$

for any x.

and definition 4.4, that

Definition. Suppose X is a random variable. The *probability distribution function* of X is given by

$$\begin{aligned} F_X(x) : \mathbb{R} &\to [0, 1] \\ x &\mapsto \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = \mathbb{P}(X \leq x) \end{aligned}$$

When dealing with discrete random variables, it was possible to only consider $\mathbb{P}(X = x)$ for some x. But now, we will have to work in more generality.

Let us observe some important properties.

1. if $x \leq y$, then $F(x) \leq F(y)$;
2. for any $a < b$,

$$\mathbb{P}(a < x \leq b) = F(b) - F(a)$$

Proof.

$$\begin{aligned} \mathbb{P}(a < X \leq b) &= \mathbb{P}(X \leq b, X > a) \\ &= \mathbb{P}(X \leq b) - \mathbb{P}(X \leq b, X \leq a) \\ &= F(b) - F(a) \end{aligned}$$

■

Lecture 15 – 25/02/26

Proposition 8.1

A probability distribution function F is right continuous, and always has left limits satisfying

$$\lim_{y \to x^-} F(y) \leq F(x)$$

Proof. Let (x_n) be a decreasing sequence satisfying $x_n \to x$ as $n \to \infty$. We want to show $F(x_n) \to F(x)$ as $n \to \infty$.

Let $A_n = \{x < X \leq x_n\}$, so $\mathbb{P}(A_n) = F(x_n) - F(x)$ by the previous. Furthermore, $A_{n+1} \subseteq A_n$ – the sequence is decreasing – so $\mathbb{P}(A_n) \to \mathbb{P}\left(\bigcap_n A_n\right)$ by continuity of probability measures. But $\bigcap_n A_n = \emptyset$, so $\mathbb{P}(A_n) \to 0$. Hence $\mathbb{P}(A_n) = F(x_n) - F(x) \to 0$ as $n \to \infty$.

Since F is increasing (and bounded), left limits always exist. That $\lim_{y \to x^-} F(y) \leq F(x)$ follows again from the fact F is increasing. ■

3. $F(x-) = \lim_{y \to x^-} F(y) = \mathbb{P}(X < x)$.

   Proof. Clearly $F(x-) = \lim_{n\to\infty} F\left(x - \frac{1}{n}\right)$. Define $B_n = \left\{X \leq x - \frac{1}{n}\right\}$, so $B_n \subseteq B_{n+1}$. Then

$$\mathbb{P}(B_n) \to \mathbb{P}\left(\bigcup_n B_n\right) = \mathbb{P}(X < x)$$

   since $\bigcup_n B_n = \{X < x\}$. ■

4. $\lim_{x\to\infty} F(x) = 1$ and $\lim_{x\to-\infty} F(x) = 0$.

Definition 8.2 (Continuous random variables)**.** A random variable X is called *continuous* if its probability distribution F is a continuous function.

Some important properties equivalent to continuity include (fixing $x \in \mathbb{R}$):

- $F(x-) = F(x)$;
- $\mathbb{P}(X < x) = \mathbb{P}(X \leq x)$;
- $\mathbb{P}(X = x) = 0$.

From now on we will consider only continuous random variables satisfying additionally that F is differentiable.

Definition 8.3 (Probability density functions)**.** The *probability density function* of a random variable X is $f(x) = F'(x)$, where F is its probability distribution.

Let us now consider some important properties of pdf's:

- $f \geq 0$;
- $\int_{-\infty}^{\infty} f(x)\,dx = 1$;

Proposition 8.4

For every $A \subseteq \mathbb{R}$,

$$\mathbb{P}(X \in A) = \int_A f(x)\,dx$$

Remark. This is a generalisation of the semi-obvious fact that

$$F(x) = \int_{-\infty}^{x} f(y)\,dy$$

What is the intuitive meaning of the probability density function? For a small Δx,

$$\mathbb{P}(x < X \le x + \Delta x) = \int_{x}^{x+\Delta x} f(y)\,dy \approx f(x)\Delta x$$

so we can think of $f(x)$ as being proportional to the probability that X is close to x.

Let us consider some important examples of continuous probability distributions.

1. *Uniform distribution* on $[a, b]$, $a < b$. We have

$$f(x) = \begin{cases} \frac{1}{b-a} & x \in [a,b] \\ 0 & \text{o/w} \end{cases}$$

$f \ge 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$, so f is indeed a pdf. Let X be a random variable with density f, denoted $X \sim [a, b]$. Then, for $x \in [a, b]$,

$$\mathbb{P}(X \le x) = \int_{-\infty}^{x} f(y)\,dy = \frac{x-a}{b-a}$$

Also, if $x > b$, $\mathbb{P}(X \le x) = 1$, and if $x < a$, $\mathbb{P}(X \le x) = 0$. A special case is the *standard uniform distribution* $X \sim U(0, 1)$

2. *Exponential distribution*. Let $\lambda > 0$, $f(x) = \lambda e^{-\lambda x}$, $x > 0$. Then

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{0}^{\infty} \lambda e^{-\lambda x}\,dx = 1$$

so f is a density. letting X have density f, we write $X \sim \text{Exp}(\lambda)$. We have

$$\begin{aligned} \mathbb{P}(X \le x) &= \int_{0}^{x} f(y)\,dy = 1 - e^{-\lambda x} \\ \mathbb{P}(X \ge x) &= e^{-\lambda x} \end{aligned}$$

Let $T \sim \text{Exp}(\lambda)$, $n \in \mathbb{N}$, $T_n = \lfloor nT \rfloor$. What is the distribution of T_n?

Let $k \in \mathbb{N}$, and let us try to evaluate $\mathbb{P}(T_n \ge k) = \mathbb{P}(nT \ge k) = \mathbb{P}(T \ge \frac{k}{n}) = e^{-\lambda \frac{k}{n}} = \left(e^{-\frac{\lambda}{n}}\right)^k$ So T_n has the geometric distribution with success parameter $p = 1 - e^{-\frac{\lambda}{n}} \approx \frac{\lambda}{n}$ for n large. In particular,

$$\frac{T_n}{n} \to T$$

as $n \to \infty$ – we can obtain the exponential random variable as a scaled limit of geometrics.

The exponential distribution has a cool property! Let $T \sim \text{Exp}(\lambda)$, and let t, $s \in \mathbb{R}^+$. Then

$$\begin{aligned} \mathbb{P}(T \ge s + t | T \ge s) &= \frac{\mathbb{P}(T \ge s + t | T \ge s)}{\mathbb{P}(T \ge s)} \\ &= \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t} \end{aligned}$$

so $\mathbb{P}(T \ge s + t | T \ge s) = \mathbb{P}(T \ge t)$. This is called the *memoryless* property (the geometric distribution also has it).

Proposition 8.5

Let T be a positive random variable which is not identically zero or infinity. Then T has the exponential distribution if and only if T has the memoryless property.

Proof. Suppose T has the memoryless property. Set $g(t) = \mathbb{P}(T \geq t)$, so

$$g(t+s) = g(t)g(s)$$

Therefore, $g(mt) = g(t)^m$ for any $t > 0$, $m \in \mathbb{N}$. Putting $t = 1$, we have $g(m) = g(1)^m$, and we will define $\lambda = -\log \mathbb{P}(T \geq 1)$ such that $g(1) = \mathbb{P}(T \geq 1) = e^{-\lambda}$. Hence $g(m) = e^{-\lambda m}$ for all $m \in \mathbb{N}$, and, for m, $n \in \mathbb{N}$, $\left(g\left(\frac{m}{n}\right)\right)^n = g(m)$, so $g(q) = e^{-\lambda q}$ for rationals q. How do we extend this to the reals?

Let $t > 0$, and let r, $s \in \mathbb{Q}^+$ such that $s < t < r$ and $|s - r| \leq \varepsilon$, which can be done by density of the rationals. Then

$$\underbrace{g(r)}_{e^{-\lambda r}} \leq g(t) \leq \underbrace{g(s)}_{e^{-\lambda s}}$$

Taking $\varepsilon \to 0$, we get $g(t) = e^{-\lambda t}$ as required. ■

Lecture 16 – 27/02/26

Definition 8.6. The *expectation* of a non-negative continuous random variable X, $X \geq 0$, with density f, is given by

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx$$

and

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f(x)\, dx$$

Definition 8.7. The expectation of a general continuous random variable X with density f is defined as follows. We put

$$X_+ = \max(X, 0)$$
$$X_- = \max(-X, 0)$$

so $X = X_+ - X_-$. If $\mathbb{E}[X_+]$ or $\mathbb{E}[X_-] < \infty$, we define

$$\mathbb{E}[X] = \mathbb{E}[X_+] - \mathbb{E}[X_-] = \int_{-\infty}^{\infty} x f(x)\, dx$$

Lemma 8.8

If X is a continuous random variable with $X \geq 0$, then

$$\mathbb{E}[X] = \int_0^{\infty} \mathbb{P}(X \geq x)\, dx$$

Proof. Observe

$$\begin{aligned}\mathbb{E}[X] &= \int_0^\infty x f(x)\,dx \\ &= \int_0^\infty \left(\int_0^x 1\,dy\right) f(x)\,dx \\ &= \int_0^\infty \left(\int_y^\infty f(x)\,dx\right) dy \\ &= \int_0^\infty \mathbb{P}(X \geq y)\,dy\end{aligned}$$

■

Remark. Recall in the discrete case we instead took expectations of the equality

$$X = \sum_{n=1}^{\infty} 1(X \geq n)$$

Can we do something similar for continuous random variables? Indeed we have

$$\begin{aligned} X(\omega) &= \int_0^\infty 1(X(\omega) \geq x)\,dx \\ \implies \mathbb{E}[X] &= \mathbb{E}\left[\int_0^\infty 1(X \geq x)\,dx\right] \\ &= \int_0^\infty \mathbb{P}(X \geq x)\,dx\end{aligned}$$

However, the interchange of expected value and integral is not justified.

Example

Let us consider some examples.

- Take $X \sim U(a,b)$. Then

$$\mathbb{E}[X] = \int_a^b x f(x)\,dx = \frac{a+b}{2}$$

- Take $X \sim \mathrm{Exp}(\lambda)$. Then

$$\mathbb{E}[X] = \int_0^\infty \lambda x e^{-\lambda x}\,dx = \frac{1}{\lambda}$$

or, perhaps more easily,

$$\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X \geq x)\,dx = \int_0^\infty e^{-\lambda x}\,dx$$

§8.1 The normal distribution

Definition 8.9 (Normal distribution)**.** Let $\mu \in \mathbb{R}$, $\sigma > 0$ be two parameters. Then

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

is the density of the *normal distribution.*

Proof. We must check f is a density. Observe

$$\begin{aligned}\int_{-\infty}^{\infty} f(x)\,dx &= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du \\ &= 2\int_{0}^{\infty} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du\end{aligned}$$

Let this integral be I. We find

$$\begin{aligned}I^2 &= \frac{2}{\pi}\int_0^\infty \int_0^\infty \exp\left(-\frac{u^2}{2}\right)\exp\left(-\frac{v^2}{2}\right) du\,dv \\ &= \frac{2}{\pi}\int_0^\infty \int_0^{\pi/2} re^{-\frac{r^2}{2}}\,d\theta dr \\ &= \int_0^\infty re^{-\frac{r^2}{2}}\,dr = 1\end{aligned}$$

Since $f(x) > 0$, we are done. ∎

Let X be a random variable with density f. We have

$$\begin{aligned}\mathbb{E}[X] &= \int_{-\infty}^{\infty} xf(x)\,dx \\ &= \underbrace{\int_{-\infty}^{\infty} (x-\mu)\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx}_{\text{odd around } \mu} + \underbrace{\int_{-\infty}^{\infty} \frac{\mu}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)}_{\mu} \\ &= \mu\end{aligned}$$

Furthermore,

$$\begin{aligned}\mathrm{Var}(X) = \mathbb{E}[(X-\mu)^2] &= \int_{-\infty}^{\infty} (x-\mu)^2 \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\ &= \sigma^2 \underbrace{\int_{-\infty}^{\infty} \frac{u^2}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du}_{\text{1 after integration by parts}} \\ &= \sigma^2\end{aligned}$$

after $u = \frac{x-\mu}{\sigma}$. So $\mathrm{Var}(X) = \sigma^2$.

When X has the normal distribution, we write $X \sim \mathcal{N}(\mu, \sigma^2)$ and we call it a normal random variable. If $X \sim \mathcal{N}(1, 0)$, X is a *standard normal variable.*

Proposition 8.10

Let X have density f. Let g be a strictly monotone function with g^{-1} differentiable. Then $g(X)$ has density given by

$$f(g^{-1}(x))\left|\frac{d}{dx}g^{-1}(x)\right|$$

Proof. Suppose g is increasing. Then

$$\mathbb{P}(g(X) \leq x) = \mathbb{P}(X \leq g^{-1}(x)) = F(g^{-1}(x))$$

Differentiating,

$$\frac{d}{dx}\mathbb{P}(g(X) \leq x) = \frac{d}{dx}\left(F(g^{-1}(x))\right) = f(g^{-1}(x))\underbrace{\frac{d}{dx}g^{-1}(x)}_{>0}$$

If g is decreasing, we put $\mathbb{P}(g(X) \leq x)$ and proceed analogously. ∎

§8.1.1 Linear transformations of a normal random variable

Let $X \sim \mathcal{N}(\mu, \sigma^2)$. Let $a \neq 0$, and $g(x) = ax + b$. What is the density of $Y = g(X)$? Clearly

$$g^{-1}(x) = \frac{x-b}{a} \implies \frac{d}{dx}g^{-1}(x) = \frac{1}{a}$$

Hence

$$\begin{aligned} f_Y(y) &= f_X(g^{-1}(y))\left|\frac{d}{dy}g^{-1}(y)\right| \\ &= \frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{(\frac{y-b}{a}-\mu)^2}{2\sigma^2}\right)\cdot\frac{1}{|a|} \\ &= \frac{1}{|a|\sigma\sqrt{2\pi}}\exp\left(-\frac{(y-(a\mu+b))^2}{2a^2\sigma^2}\right) \end{aligned}$$

so $Y \sim \mathcal{N}(a\mu + b, (a\sigma)^2)$. Hence, letting $X \sim \mathcal{N}(\mu, \sigma)^2$, we have

$$\frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1)$$

This is a very helpful result! Say, for example, I have $X \sim \mathcal{N}(\mu, \sigma^2)$, and I want to find $\mathbb{P}(-2\sigma < X - \mu < 2\sigma)$. This is just $\mathbb{P}(-2 < \mathcal{N}(0,1) < 2)$ for the standard normal variable $\mathcal{N}(0,1)$.

Definition 8.11 (Standard normal probability distribution function)**.** Let

$$\Phi(x) = \int_{-\infty}^{x}\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{u^2}{2}\right) = \mathbb{P}(\mathcal{N}(0,1) \leq x)$$

Now

$$\phi(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$$

so $\phi(x) - \phi(-x) = 0$, and hence $(\Phi(x) + \Phi(-x))' = 0$, so $\Phi(x) + \Phi(-x)$ takes a constant value of 1.

Definition 8.12 (Median)**.** Let X be a continuous random variable. The *median* of X, denoted by m, is the number satisfying

$$\mathbb{P}(X \geq m) = \mathbb{P}(X \leq m) = \frac{1}{2}$$

or

$$\int_{-\infty}^{m} f(x)\,dx = \int_{m}^{\infty} f(x)\,dx = \frac{1}{2}$$

Example

Suppose $X \sim \mathcal{N}(\mu, \sigma^2)$. We claim the median is μ. Observe

$$\mathbb{P}(X \le \mu) = \mathbb{P}\left(\frac{X-\mu}{\sigma} \le 0\right) = \mathbb{P}(\mathcal{N}(0,1) \le 0) = \frac{1}{2}$$

§8.2 Multivariate density functions

In the discrete case, we defined the joint distribution of $X_1, \dots, X_n$ in terms of

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n)$$

However, in the continuous case, such a probability is of course zero, so we cannot talk about this. Instead, we say

Definition 8.13 (Joint probability distribution function)**.** The *probability distribution function* F for a vector $X = (X_1, \dots, X_n)$, where X_i are random variables, is such that

$$F(x_1, \dots, x_n) = \mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n)$$

Definition 8.14 (Joint probability density function)**.** The *probability density function* f for a vector $X = (X_1, \dots, X_n)$, where X_i are random variables, is such that

$$f(x_1, \dots, x_n) = \frac{\partial^n F}{\partial x_1 \dots \partial x_n}(x_1, \dots, x_n)$$

or, equivalently,

$$\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f(y_1, \dots, y_n)\, dy_n \dots dy_1$$

Proposition 8.15

Let $B \subset \mathbb{R}^n$. Then

$$\mathbb{P}((X_1, \dots, X_n) \in B) = \int_B f(y_1, \dots, y_n)\, dy_1 \dots dy_n$$

Definition 8.16 (Independence)**.** Let $X_1, \dots, X_n$ be random variables. We say they are independent if, for any $x_1, \dots, x_n$,

$$\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) = \mathbb{P}(X_1 \le x_1) \dots \mathbb{P}(X_n \le x_n)$$

Proposition 8.17

Suppose $X = (X_1, \dots, X_n)$ has density f. Then

- Suppose $X_1, \dots X_n$ are independent and have densities given by $f_1, \dots, f_n$. Then $f(x_1, \dots, x_n) = f_1(x_1) \dots f_n(x_n)$.
- Suppose $f(x_1, \dots, x_n) = f_1(x_1) \dots f_n(x_n)$ for some non-negative functions $f_1, \dots, f_n$. Then $X_1, \dots, X_n$ are independent with densities proportional to each of $f_1, \dots, f_n$.

Proof.

- We have

$$\begin{aligned}\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) &= \mathbb{P}(X_1 \le x_1) \dots \mathbb{P}(X_n \le x_n) \\ &= \left(\int_{-\infty}^{x_1} f_1(y_1)\, dy_1\right) \dots \left(\int_{-\infty}^{x_n} f_n(y_n)\right) \\ &= \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_n} f_1(y_1) \dots f_n(y_n)\, dy_n \dots dy_1\end{aligned}$$

so $f(x_1, \dots, x_n) = f_1(x_1) \dots f_n(x_n)$ as required.

- We have

$$\begin{aligned}\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) &= \int_{-\infty}^{x_1} \dots \int_{-\infty}^{x_n} f(y_1, \dots, y_n)\, dy_n \dots dy_1 \\ &= \left(\int_{-\infty}^{x_1} f_1(y_1)\, dy_1\right) \dots \left(\int_{-\infty}^{x_n} f_n(y_n)\, dy_n\right)\end{aligned}$$

Observe $\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f_1(y_1) \dots f_n(y_n)\, dx_1 \dots dx_n = 1$. Hence

$$\mathbb{P}(X_1 \le x_1, \dots, X_n \le x_n) = \frac{\int_{-\infty}^{x_1} f_1(y_1)\, dy_1}{\int_{-\infty}^{\infty} f_1(y_1)\, dy_1} \cdots \frac{\int_{-\infty}^{x_n} f_n(y_n)\, dy_n}{\int_{-\infty}^{\infty} f_n(y_n)\, dy_n}$$

So $X_1, \dots, X_n$ are independent, with X_i having density

$$\frac{f(x_i)}{\int_{-\infty}^{\infty} f(y_i)\, dy_i}$$

■

§8.2.1 Marginal density

Definition 8.18 (Marginal density)**.** Let $X = (X_1, \dots, X_n)$ be a vector of random variables. The *marginal density* of X_i, $f_{X_i}(x_i)$, is the density of X_i, where all other variables' values are ignored.

Proposition 8.19

Let $X = (X_1, \dots, X_n)$ be a vector of random variables. The marginal density is given by

$$f_{X_i}(x_i) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, \dots, x_i, \dots, x_n)\, dx_n \dots dx_{i+1}\, dx_{i-1} \dots dx_1$$

Proof. Without loss of generality take $i = 1$. Then

$$\begin{aligned}\mathbb{P}(X_1 \le x) &= \mathbb{P}(X_1 \le x, X_2 \in \mathbb{R}, \dots, X_n \in \mathbb{R}) \\ &= \int_{-\infty}^{x} \left(\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, \dots, x_n)\, dx_n \dots dx_2\right) dx_1\end{aligned}$$

so the expression inside the integral must be the density of X_1. ■

§8.2.2 Sum of independent random variables

Proposition 8.20

Suppose X and Y are independent random variables with density f_X and f_Y respectively. The density of $X+Y$ is given by

$$f_{X+Y}(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$$

Proof. Observe

$$\begin{aligned}
\mathbb{P}(X+Y \leq z) &= \mathbb{P}((X,Y) \in \{(x,y) \in \mathbb{R}^2 : x+y \leq z\}) \\
&= \int_{x+y\leq z} f_{X,Y}(x,y)\, dx\, dy \\
&= \int_{x+y\leq z} f_X(x) f_Y(y)\, dx\, dy \\
&= \int_{-\infty}^{\infty} f_X(x) \int_{-\infty}^{z-x} f_Y(y)\, dy \\
&= \int_{-\infty}^{\infty} f_X(x) \int_{-\infty}^{z} f_Y(y-x)\, dy\, dx \\
&= \int_{-\infty}^{z} \left(\int_{-\infty}^{\infty} f_X(x) f_Y(y-x)\, dx \right) dy
\end{aligned}$$

so the density of $X+Y$ is as required. ∎

§8.2.3 Conditional density

Definition 8.21 (Conditional density)**.** Let X and Y be random variables with joint density $f_{X,Y}$ and marginal densities f_X and f_Y. The *conditional density* of X given $Y=y$ is

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

for $f_Y(y) > 0$.

Proposition 8.22 (Law of total probability for continuous random variables)

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dy = \int_{-\infty}^{\infty} f_{X|Y}(x|y) f_Y(y)\, dy$$

Definition 8.23 (Conditional expectation for continuous random variables)**.** We define $\mathbb{E}[X|Y]$, the *conditional expectation of X given y*, to be $g(Y)$, where

$$g(y) = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\, dx$$

Theorem 8.24

Let X be a random variable with values in $D \subseteq \mathbb{R}^n$ and density f. Let g be a bijection from D to $g(D)$ with continuous derivatives on D and $|J| \neq 0$ for any $x \in D$, where $|J|$ is the Jacobian of g^{-1}. Then, the random variable $Y = g(X)$ has density

$$f_Y(y) = f_X\left(g^{-1}(y)\right)|J|$$

Example

Suppose X, Y are independent and $X, Y \sim \mathcal{N}(0,1)$. What are the distributions of the magnitude and argument of $X + iY$? By the above,

$$\begin{aligned} f_{R,\theta}(r,\theta) &= f_{X,Y}(r\cos\theta, r\sin\theta)|J| = f_X(r\cos\theta)f_Y(r\sin\theta)r \\ &= \frac{1}{\sqrt{2\pi}}e^{-\frac{r^2\cos^2\theta}{2}} \cdot \frac{1}{\sqrt{2\pi}}e^{-\frac{r^2\sin^2\theta}{2}}r \\ &= \frac{1}{2\pi}re^{-\frac{r^2}{2}} \end{aligned}$$

Hence, $\theta \sim U(0, 2\pi)$, R has density $re^{-\frac{r^2}{2}}$, and they are independent.

§8.3 Order statistics of a random sample

Let $X_1, \ldots, X_n$ be i.i.d. random variables with probability distribution function F and density f. For a given sample, let $Y_1 \leq \cdots \leq Y_n$ be the X_i ordered from smallest to largest – called the order statistics – and let us try to find the density of $(Y_1, \ldots, Y_n)$.

Firstly, observe

$$\mathbb{P}(Y_1 \leq x) = 1 - \mathbb{P}(Y_1 > x) = 1 - \mathbb{P}(X_1 > x)^n = 1 - (1 - F(x))^n$$

by independence. Differentiating, we obtain

$$f_{Y_1}(x) = \frac{d}{dx}\mathbb{P}(Y_1 \leq x) = nf(x)(1 - F(x))^{n-1}$$

Similarly,

$$\begin{aligned} &\mathbb{P}(Y_n \leq x) = F(x)^n \\ \implies\ & f_{Y_n}(x) = nf(x)F(x)^{n-1} \end{aligned}$$

Can we generalise these results? Yes!

Proposition 8.25

Let $X_1, \ldots, X_n$ be i.i.d. random variables with probability distribution function F and density f, and let $Y_1, \ldots, Y_n$ be their order statistics. Then

$$f_{(Y_1,\ldots,Y_n)}(x_1, \ldots, x_n) = \begin{cases} n!f(x_1)\ldots f(x_n) & x_1 < \cdots < x_n \\ 0 & \text{o/w} \end{cases}$$

Proof. Considering all permutations of the X_i which give the same ordered Y_i's, we have

$$\begin{aligned}
&\mathbb{P}(Y_1 \leq x_1, \ldots, Y_n \leq x_n) \\
&= n! \cdot \mathbb{P}(X_1 \leq x_1, \ldots, X_n \leq x_n, X_1 \leq \cdots \leq X_n) \\
&= n! \int_{-\infty}^{x_1} \int_{u_1}^{x_2} \cdots \int_{u_{n=1}}^{x_n} f(u_n) f(u_{n-1}) \ldots f(u_1)\, du_n \ldots du_1 \\
\implies \; & f_{(Y_1,\ldots,Y_n)}(x_1, \ldots, x_n) = \begin{cases} n! f(x_1) \ldots f(x_n) & x_1 < \cdots < x_n \\ 0 & \text{o/w} \end{cases}
\end{aligned}$$

after differentiation and the application of FTC. Notably, the Y_i are not independent, since the joint distribution only factorises in the case $x_1 < \cdots < x_n$. ■

Example

Let X_i be independent with $X_i \sim \text{Exp}(\lambda_i)$, and let $X = \min(X_1, \ldots, X_n)$. Then

$$\begin{aligned}
\mathbb{P}(X \leq x) &= 1 - \left(1 - \left(1 - e^{-\lambda_1 x}\right)\right) \cdots \left(1 - \left(1 - e^{-\lambda_n x}\right)\right) \\
&= 1 - e^{-(\lambda_1 + \cdots + \lambda_n)x}
\end{aligned}$$

so $X \sim \text{Exp}\left(\sum_{i=1}^n \lambda_i\right)$. Now let $Y_1, \ldots, Y_n$ be the order statistics, and set $Z_1 = Y_1$, $Z_2 = Y_2 - Y_1$, $Z_i = Y_i - Y_{i-1}$ for $i \leq n$ – so the Z_i's represent the spacing between the samples. What is the joint density of $(Z_1, \ldots, Z_n)$?

Observe

$$\underbrace{\begin{pmatrix} Z_1 \\ \vdots \\ Z_n \end{pmatrix}}_{\mathbf{Z}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & 0 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}}_{\mathbf{Y}}$$

and also that

$$y_j = \sum_{i=1}^{j} z_i$$

Now, by the theorem,

$$\begin{aligned}
f_{(Z_1,\ldots,Z_n)}(z_1, \ldots, z_n) &= f_{(Y_1,\ldots,Y_n)}(y_1, \ldots, y_n)|J| \\
&= n! f(y_1) \cdots f(y_n) \\
&= n! \lambda e^{-\lambda z_1} \lambda e^{-\lambda(z_1+z_2)} \ldots \lambda e^{-\lambda(z_1+\cdots+z_n)} \\
&= \prod_{i=1}^{n} \left((n-i+1)\lambda e^{-\lambda(n-i+1)z_i}\right)
\end{aligned}$$

so the Z_i's are independent, and $Z_i \sim \text{Exp}(\lambda(n-i+1))$.

§9 Moment generating functions

Definition 9.1 (Moment generating functions)**.** Let X be a random variable with density f. The *moment generating function* or *mgf* of X is defined to be

$$m(\theta) = \mathbb{E}[e^{\theta X}] = \int_{-\infty}^{\infty} e^{\theta x} f(x)\, dx$$

whenever this integral is finite.

Theorem 9.2

The mgf uniquely determines the distribution of a random variable, provided it is defined for an open interval of values of θ. Furthermore,

$$m^{(r)}(\theta) = \mathbb{E}[X^r]$$

Example

The *gamma distribution*, for $n \in \mathbb{N}$ and $\lambda > 0$, has density

$$f(x) = \frac{e^{-\lambda x}\lambda^n x^{n-1}}{(n-1)!}$$

which has the distribution of a sum of n i.i.d. exponential random variables. Let us check that it is a density.

$$\begin{aligned} I_n = \int_0^\infty f(x)\, dx &= \int_0^\infty \lambda e^{-\lambda x} \cdot \frac{\lambda^{n-1}x^{n-1}}{(n-1)!} \\ &= \int_0^\infty \lambda e^{-\lambda x} \cdot \frac{\lambda^{n-2}x^{n-2}}{(n-2)!} && \text{(parts)} \\ &= I_{n-1} = \cdots = I_1 = 1 \end{aligned}$$

Suppose $X \sim \Gamma(n, \lambda)$. What is the mgf of X? We have

$$\begin{aligned} m(\theta) &= \int_0^\infty e^{\theta x} e^{-\lambda x} \frac{\lambda^n x^{n-1}}{(n-1)!}\, dx \\ &= \frac{\lambda^n}{(\lambda-\theta)^n} \int_0^\infty e^{-(\lambda-\theta)x} \frac{(\lambda-\theta)^n x^{n-1}}{(n-1)!}\, dx \end{aligned}$$

If $\theta < \lambda$, then

$$m(\theta) = \left(\frac{\lambda}{\lambda - \theta}\right)^n$$

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Proposition 9.3

If $X_1, \dots, X_n$ are independent, then $X_1 + \cdots + X_n$ has mgf

$$m(\theta) = \mathbb{E}[e^{\theta(X_1+\cdots+X_n)}] = \prod_{i=1}^{n} \mathbb{E}[e^{\theta X_i}]$$

Example (Sum of gamma random variables)

Suppose $X \sim \Gamma(n, \lambda)$ and $Y \sim \Gamma(m, \lambda)$, and $X \perp Y$. Then

$$\mathbb{E}[e^{\theta(X+Y)}] = \mathbb{E}[e^{\theta X}]\mathbb{E}[e^{\theta Y}] = \left(\frac{\lambda}{\lambda-\theta}\right)^n \left(\frac{\lambda}{\lambda-\theta}\right)^m = \left(\frac{\lambda}{\lambda-\theta}\right)^{n+m}$$

so $X + Y \sim \Gamma(n+m, \lambda)$. It is a consequence of this that, if $X_1, \dots, X_n$ are i.i.d. $\text{Exp}(\lambda)$ random variables, then $X_1 + \cdots + X_n \sim \Gamma(n, \lambda)$.

Remark. One can generalise the gamma distribution to non-integer m by replacing $(n-1)!$ with

$$\Gamma(\alpha) = \int_0^\infty e^{-x} x^{\alpha-1} \, dx$$

so that $X \sim \Gamma(\alpha, \lambda)$ if

$$f(x) = \frac{e^{-\lambda x} \lambda^\alpha x^{\alpha-1}}{\Gamma(\alpha)}$$

Definition 9.4 (Cauchy distribution)**.** The *Cauchy distribution* has density

$$f(x) = \frac{1}{\pi(1+x^2)}$$

Hence, the mgf is

$$m(\theta) = \int_{-\infty}^{\infty} \frac{e^{\theta x}}{\pi(1+x^2)} \, dx = \infty$$

Suppose $X \sim f$, then observe X, $2X, \dots$ all have the same mgf, but not the same distribution. Hence mgfs do not uniquely determine distributions unless they are finite.

What is the mgf of a normally distributed random variable? We have

$$\begin{aligned}
m(\theta) &= \int_{-\infty}^{\infty} e^{\theta x} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{x^2}{2\sigma^2} + 2\frac{x}{2\sigma^2}\left(\mu + \theta\sigma^2\right) - \frac{\mu^2}{2\sigma^2}\right) dx \\
&= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}\left(x - (\mu+\theta\sigma^2)\right)^2 + \mu\theta + \frac{\theta^2\sigma^2}{2}\right) dx \\
&= \exp\left(\mu\theta + \frac{\theta^2\sigma^2}{2}\right) \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2\sigma^2}\left(x - (\mu+\theta\sigma^2)\right)^2\right) dx
\end{aligned}$$

Then, the integrand is the density of $\mathcal{N}(\mu + \theta\sigma^2, \sigma^2)$, so we obtain

$$m(\theta) = \exp\left(\mu\theta + \frac{\theta^2\sigma^2}{2}\right)$$

This gives an easy way to see that

Proposition 9.5

If $X \sim \mathcal{N}(\mu, \sigma^2)$, $Y \sim \mathcal{N}(\nu, \tau^2)$ and $X \perp Y$, then $X + Y \sim \mathcal{N}(\mu + \nu, \sigma^2 + \tau^2)$.

§9.1 Multivariate moment generating functions

Definition 9.6 (Multivariate moment generating functions)**.** Let $X = (X_1, \ldots, X_n)$ be a random variable. The mgf of X is defined to be

$$m(\theta) = \mathbb{E}[e^{\theta^\intercal X}]$$

Theorem 9.7

If the mgf is finite for an open set of values of θ, then it uniquely determines the distribution – furthermore,

$$\left.\frac{\partial^r m}{\partial \theta_i^r}\right|_{\theta=0} = \mathbb{E}[X_i^r], \qquad \left.\frac{\partial^{r+s} m}{\partial \theta_i^r \partial \theta_j^s}\right|_{\theta=0} = \mathbb{E}[X_i^r X_j^s]$$

Proposition 9.8

Let $(X_1, \ldots, X_n) = X$ be a random variable in $\mathbb{R}^n$. Then

$$m(\theta) = \prod_{i=1}^n \mathbb{E}[e^{\theta_i X_i}]$$

if and only if $X_1, \ldots, X_n$ are independent.

Proof. If $X_1, \ldots, X_n$ are independent, then $m(\theta)$ factorises. Conversely, if $m(\theta)$ factorises, then the X_i must be independent by the uniqueness of mgfs. ∎

§9.2 Multivariate Gaussian random variables

Definition 9.9 (Multivariate Gaussian random variables)**.** Let $X = (X_1, \ldots, X_n)$ be a random variable. We say X is *Gaussian* if, for any $u \in \mathbb{R}^n$, $u^\intercal X$ is a Gaussian random variable in $\mathbb{R}$.

Proposition 9.10

Let $X \in \mathbb{R}^n$ be a Gaussian vector. Let A be an m by n matrix and let $b \in \mathbb{R}^m$. Then $AX + b$ is also Gaussian.

Proof. Let $u \in \mathbb{R}^m$ – we want to show $u^\intercal(AX+b)$ is Gaussian in $\mathbb{R}$. Observe

$$\begin{aligned} u^\intercal(AX+b) &= u^\intercal AX + u^\intercal b \\ &= v^\intercal X + u^\intercal b \end{aligned}$$

where $v^\intercal X$ is Gaussian (since X is) and $u^\intercal b$ is constant. The result follows. ■

Definition 9.11 (Multivariate mean and variance)**.** A random vector variable X has mean

$$\mu = \mathbb{E}[X] = \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix}$$

and variance

$$V = \mathrm{Var}(X) = \mathbb{E}[(X-\mu)\cdot(X-\mu)^\intercal]$$

which is an n-by-n symmetric matrix, where expectation is taken pointwise.

> **Remark.** Observe that $\mathrm{Var}(X)_{i,j} = \mathrm{Cov}(X_i, X_j)$.

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What is the expectation of $u^\intercal X$? It should be

$$\mathbb{E}\left[\sum_{i=1}^n u_i X_i\right] = \sum_{i=1}^n u_i \mathbb{E}[X_i] = u^\intercal \mu$$

Furthermore,

$$\mathrm{Var}(u^\intercal X) = \mathrm{Var}\left(\sum_{i=1}^n u_i X_i\right) = \sum_{i,j=1}^n u_i \,\mathrm{Cov}(X_i, X_j) u_j = u^\intercal V u$$

Proposition 9.12

V is a non-negative definite matrix – for any $u \in \mathbb{R}^n$,

$$u^\intercal V u \geq 0$$

Proof. $u^\intercal V u = \mathrm{Var}(u^\intercal X) \geq 0$. ■

Let us also find the mgf. We have, for any vector λ, $\lambda^\intercal X \sim \mathcal{N}(\lambda^\intercal \mu, \lambda^\intercal V \lambda)$. It follows immediately from the expression for the mgf of a Gaussian random variable in $\mathbb{R}$ that

$$m(\lambda) = \exp\left(\lambda^\intercal \mu + \frac{\lambda^\intercal V \lambda}{2}\right)$$

Recall that the mgf uniquely defines the distribution in any case – hence, for a Gaussian random variable in particular, we conclude μ and V are sufficient.

Lemma 9.13 (Construction of a Gaussian random variable)
Let $Z_1, \dots, Z_n$ be i.i.d $\mathcal{N}(0,1)$ random variables. Then $Z = (Z_1, \dots, Z_n)$ is a Gaussian vector.

Proof. Let $u \in \mathbb{R}^n$, and we need to show $u^\intercal Z$ is normal in $\mathbb{R}$. We have that the mgf of $u^\intercal Z$ is

$$\begin{aligned} m(\lambda) &= \mathbb{E}\left[e^{\lambda \sum_{i=1}^n u_i Z_i}\right] \\ &= \prod_{i=1}^n \mathbb{E}\left[e^{\lambda u_i z_i}\right] \\ &= \prod_{i=1}^n \exp\left(\frac{\lambda^2 u_i^2}{2}\right) \\ &= \exp\left(\frac{\lambda^2 |u|^2}{2}\right) \end{aligned}$$

Hence, $u^\intercal Z \sim \mathcal{N}(0, |u|^2)$. ∎

> **Remark.** For this vector, we have in particular $\mu = \mathbf{0}$, $V = I_n$ – we write $Z \sim \mathcal{N}(0, I_n)$.

Let $\mu \in \mathbb{R}^n$ and let V be a non-negative definite matrix. We want to construct a Gaussian vector with mean μ and variance V, using the standard Gaussian Z. Recall that, for $n = 1$, we can do this by taking $X = \mu + \sigma Z$. However, for this, we need to somehow take the square root of the variance – how will this generalise?

Definition 9.14 (Matrix square root)**.** Let V be a non-negative definite matrix such that

$$V = U^\intercal D U$$

where $U^\intercal = U^{-1}$ and D is a diagonal matrix, which has diagonal entries all ≥ 0. Then

$$\sqrt{V} = U^\intercal \sqrt{D} U$$

where the square root of D is produced by taking the square root of each of its diagonal entries.

> **Remark.** Reassuringly, this definition satisfies $(\sqrt{V})^2 = V$.

Lemma 9.15 (Construction of a general Gaussian random variable)
Let $\mu \in \mathbb{R}^n$ and V be a non-negative definite matrix. Let $Z_1, \dots, Z_n$ be i.i.d $\mathcal{N}(0,1)$ random variables and $Z = (Z_1, \dots, Z_n)$. Let $X = \mu + \sigma Z$, where σ is the square root of V. Then X is Gaussian with mean μ and variance $V = \sigma^2$, denoted $X \sim \mathcal{N}(\mu, V)$.

Proof. X is Gaussian by proposition 9.10. Clearly $\mathbb{E}[X] = \mu$, and

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mu)(X - \mu)^\intercal]$$

$$
\begin{aligned}
&= \mathbb{E}[(\sigma Z)(Z^\intercal \sigma^\intercal)] \\
&= \sigma \mathbb{E}[Z Z^\intercal] \sigma^\intercal \\
&= \sigma I_n \sigma^\intercal \\
&= V
\end{aligned}
$$

■

Proposition 9.16

Let $X \sim \mathcal{N}(\mu, V)$, where V is positive definite. Then, the density of X is given by

$$f_X(x) = \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\frac{(x-\mu)^\intercal V^{-1}(x-\mu)}{2}\right)$$

Proof. We can write $X = \mu + \sigma Z$, where $Z \sim \mathcal{N}(0, I_n)$, and $Z = \sigma^{-1}(x - \mu)$. Hence, by theorem 8.24, we have

$$
\begin{aligned}
f_X(x) &= f_Z(z)|J| \\
&= \frac{1}{(2\pi)^{n/2}} \exp\left(-\frac{z^\intercal z}{2}\right) \frac{1}{\det \sigma} \\
&= \frac{1}{(2\pi)^{n/2} \det \sigma} \exp\left(-\frac{(x-\mu)^\intercal (\sigma^{-1})^\intercal \sigma^{-1}(x-\mu)}{2}\right) \\
&= \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\frac{(x-\mu)^\intercal V^{-1}(x-\mu)}{2}\right)
\end{aligned}
$$

■

We now seek to generalise this result to non-negative definite variance. If V has some eigenvalue equal to zero, there exists an orthogonal change of basis such that

$$V = \begin{pmatrix} U & 0 \\ 0 & 0 \end{pmatrix}$$

where U is an m by m positive definite matrix with all other entries filled in by zero. Then, we can write $\mu = (\lambda, \nu)$ where $\lambda \in \mathbb{R}^m$, $\nu \in \mathbb{R}^{n-m}$, so that $X = (y, \nu)$ for a constant vector ν and $Y \sim \mathcal{N}(\lambda, U)$ with density

$$f_Y(y) = \frac{1}{\sqrt{(2\pi)^m \det U}} \exp\left(-\frac{(y-\lambda)^\intercal U^{-1}(y-\lambda)}{2}\right)$$

Proposition 9.17

Let $X = (X_1, \dots, X_n)$ be a Gaussian vector, $\mu = \mathbb{E}[X]$, $V = \mathrm{Var}(X)$ positive definite. Then $X_1, \dots, X_n$ are independent if and only if V is diagonal.

The forward implication is true for any random variable. We give two proofs of the converse.

Proof 1. Let

$$V = \begin{pmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{pmatrix}$$

Let $\mu = (\mu_1, \ldots, \mu_n)$, so that

$$\begin{aligned} f_X(x) &= \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\frac{(x-\mu)^\intercal V^{-1}(x-\mu)}{2}\right) \\ &= \frac{1}{\sqrt{(2\pi)^n \det V}} \exp\left(-\sum_{i=1}^n \frac{(x_i-\mu_i)^2}{2\lambda_i}\right) \\ &= \prod_{i=1}^n \frac{1}{\lambda_i\sqrt{2\pi}} \exp\left(-\frac{(x_i-\mu_i)^2}{2\lambda_i}\right) \end{aligned}$$

so X_i are independent as required, and furthermore each is normal with mean μ_i and variance λ_i. ■

Proof 2. Observe

$$\begin{aligned} m(\theta) = \mathbb{E}[e^{\theta^\intercal X}] &= \exp\left(\theta^\intercal \mu + \frac{\theta^\intercal V \theta}{2}\right) \\ &= \exp\left(\sum_{i=1}^n \theta_i \mu_i + \sum_{i=1}^n \frac{\theta_i^2 \lambda_i}{2}\right) \\ &= \prod_{i=1}^n \exp\left(\theta_i \mu_i + \frac{\theta_i^2 \lambda_i}{2}\right) \end{aligned}$$

so the mgf factorises and hence the X_i are independent. ■

§9.3 Bivariate Gaussians

Definition 9.18 (Correlation)**.** Let $X = (X_1, X_2)$ be a random variable in $\mathbb{R}^2$, with $\mu_i = \mathbb{E}[X_i]$, $\sigma_i^2 = \operatorname{Var}(X_i)$. Then the *correlation* is given by

$$\rho = \operatorname{Corr}(X_1, X_2) = \frac{\operatorname{Cov}(X_1, X_2)}{\sqrt{\operatorname{Var}(X_1)\operatorname{Var}(X_2)}}$$

Lemma 9.19

For any X, $-1 \le \rho \le 1$.

Proof. Cauchy-Schwarz. ■

Lemma 9.20

The variance matrix of X, which is given by

$$V = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

is positive semi-definite.

Proof. Let $u = (u_1, u_2)^\intercal$, so

$$u^\intercal V u = (1 \pm \rho)(\sigma_1^2 u_1^2 + \sigma_2^2 u_2^2) \mp \rho(\sigma_1 u_1 - \sigma_2 u_2)^2$$

Depending on whether $\rho \in [-1, 0]$ or $\rho \in [0, 1]$, we take the first plus sign or the first minus sign, respectively. ∎

Proposition 9.21 (Conditional expectation for bivariate Gaussians)

Suppose X is a bivariate Gaussian, then

$$\mathbb{E}[X_2 \mid X_1] = \mu_2 + a(X_1 - \mu_1)$$

where $a = \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_1)}$.

Proof. Let $Y = X_2 - aX_1$. Observe

$$\begin{pmatrix} X_1 \\ Y \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -a & 1 \end{pmatrix} \begin{pmatrix} X_1 \\ X_2 \end{pmatrix}$$

so (X_1, Y) is Gaussian. Observe

$$\begin{aligned} \mathrm{Cov}(X_1, Y) &= \mathrm{Cov}(X_1, X_2 - aX_1) \\ &= \mathrm{Cov}(X_1, X_2) - a\,\mathrm{Var}(X_1) \end{aligned}$$

Taking $a = \frac{\mathrm{Cov}(X_1, X_2)}{\mathrm{Var}(X_1)}$ gives $\mathrm{Cov}(X_1, Y) = 0$, so X_1 and Y are independent. Then

$$\begin{aligned} \mathbb{E}[X_2 | X_1] &= \mathbb{E}[X_2 - aX_1 | X_1] + \mathbb{E}[aX_1 | X_1] \\ &= \mathbb{E}[X_2 - aX_1] + aX_1 \\ &= \mu_2 - a\mu_1 + aX_1 \end{aligned}$$

∎

§10 Convergence results and limit theorems

Definition 10.1 (Convergence in probability)**.** A sequence of random variables (X_n) converges to X *in probability*, $X_n \xrightarrow{\mathbb{P}} X$ if, for any $\varepsilon > 0$,

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0$$

as $n \to \infty$.

Theorem 10.2 (Weak law of large numbers)

Let (X_n) be a sequence of i.i.d. random variables with finite mean. Let $S_n = X_1 + \cdots + X_n$, then

$$\frac{S_n}{n} \xrightarrow{\mathbb{P}} \mu$$

as $n \to \infty$.

Proof. Assume $\sigma^2 = \mathrm{Var}(X_1) < \infty$. Observe

$$\begin{aligned}\mathbb{P}\left(\left|\frac{S_n}{n} - \mu\right| > \varepsilon\right) &= \mathbb{P}(|S_n - n\mu| > n\varepsilon) \\ &\leq \frac{\mathrm{Var}(S_n)}{n^2\varepsilon^2} \\ &= \frac{\sigma^2}{n\varepsilon^2} \to 0\end{aligned}$$

as $n \to \infty$. ∎

Definition 10.3 (Almost sure/probability 1 convergence)**.** Let (X_n) be a sequence of random variables and X a random variable. We say that $X_n \to X$ *almost surely* or *with probability 1* if

$$\mathbb{P}\left(\lim_{n\to\infty} X_n = X\right) = 1$$

Perhaps it is most helpful to return to the definition of random variables. We have that there exists $A \in \mathcal{F}$ such that $\mathbb{P}(A) = 1$ and, for ω in that set, the functions X_n converge pointwise to the function X.

Proposition 10.4

If $X_n \to 0$ a.s. as $n \to \infty$, then $X_n \xrightarrow{\mathbb{P}} 0$ as $n \to \infty$.

Proof. Given an $\varepsilon > 0$, we want to show that $\mathbb{P}(|X_n| \leq \varepsilon) \to 1$ as $n \to \infty$. Observe

$$\mathbb{P}(|X_n| \leq \varepsilon) \geq \mathbb{P}\left(\underbrace{\bigcap_{m=n}^{\infty} |X_m| \leq \varepsilon}_{A_n}\right)$$

Then the events A_n are increasing, that is $A_n \subseteq A_{n+1}$. It follows by continuity of probability measures that $\mathbb{P}(A_n)$ is increasing and converges to $\mathbb{P}\left(\bigcup_n A_n\right)$ as $n \to \infty$. Hence

$$\mathbb{P}(A_n) \to \mathbb{P}\left(\bigcup_n \bigcap_{m=n}^{\infty} |X_m| \leq \varepsilon\right)$$

so that

$$\lim_{n\to\infty} \mathbb{P}(|X_n| \leq \varepsilon) \geq \mathbb{P}(\exists n \text{ s.t. } m \geq n \implies |X_m| < \varepsilon)$$

$$\geq \mathbb{P}(\forall \varepsilon > 0, \exists n \text{ s.t. } m \geq n \implies |X_m| < \varepsilon)$$

But this is exactly the set of ω such that X_n converges pointwise to X, so it has probability 1 by assumption. Hence

$$\begin{aligned} &\lim_{n\to\infty} \mathbb{P}(|X_n| \leq \varepsilon) \geq 1 \\ \implies &\lim_{n\to\infty} \mathbb{P}(|X_n| \leq \varepsilon) = 1 \end{aligned}$$

as required. ■

> **Remark.** In fact convergence in probability does not imply almost sure convergence, as one can show by considering a well-chosen counterexample.

Theorem 10.5 (Strong law of large numbers)
Let (X_n) be a sequence of i.i.d. random variables with finite mean. Let $S_n = X_1 + \cdots + X_n$, then

$$\frac{S_n}{n} \to \mu$$

a.s. as $n \to \infty$.

Proof (non-examinable). Assume $\sigma^2 = \mathrm{Var}(X_1) < \infty$ and $\mathbb{E}[X_1^4] < \infty$. Let $Y_i = X_i - \mu$, so $\mathbb{E}[Y_i] = 0$, and

$$\mathbb{E}[Y_1^4] = \mathbb{E}[(X_1 - \mu)^4] \leq \mathbb{E}[(2\max(\mathbb{E}[X_1], \mu))^4] \leq 2^4 \left(\mathbb{E}[X_1^4] + \mu^4\right)$$

Set now $S_n = \sum_{i=1}^n Y_i$, and we must show $S_n/n \to 0$ a.s. as $n \to \infty$.

We claim that

$$\mathbb{P}\left(\sum_{n=1}^{\infty} \left(\frac{S_n}{n}\right)^4 < \infty\right) = 1$$

It suffices to show that

$$\mathbb{E}\left[\sum \left(\frac{S_n}{n}\right)^4\right] < \infty$$

Observe

$$\begin{aligned} \mathbb{E}\left[\sum \left(\frac{S_n}{n}\right)^4\right] &= \sum \frac{1}{n^4}\mathbb{E}[S_n^4] \\ &= \sum \frac{1}{n^4}\mathbb{E}\left[\sum_{i=1}^n Y_i^4 + \binom{4}{2} \sum_{1\leq i<j\leq n} Y_i^2 Y_j^2 + R\right] \end{aligned}$$

where R is a sum of terms of the form

$$Y_i^3 Y_j,\ Y_i^2 Y_j Y_k,\ Y_i Y_j Y_k Y_l$$

for distinct indices i, j, k, l. For such a term, the expectation factorises by independence, and then, since $\mathbb{E}[X_i] = 0$ and a first power appears in each of them, we obtain 0.

Let us consider the $X_i^2 X_j^2$ terms. We have

$$\mathbb{E}[X_i^2 X_j^2] = \mathbb{E}[Y_i^2]^2$$

Now, by Jensen with $f(x) = x^2$ or just non-negativity of variance, we have $\mathbb{E}[Y_i^2]^2 \le \mathbb{E}[Y_i^4]$, which is finite. Now Furthermore,

$$\begin{aligned}\sum \frac{1}{n^4}\mathbb{E}\left[\sum_{i=1}^{n} Y_i^4 + \binom{4}{2} \sum_{1\le i<j\le n} Y_i^2 Y_j^2\right] &\le \sum \frac{1}{n^4}\left(n\mathbb{E}[Y_1^4] + 6\frac{n(n-1)}{2}\mathbb{E}[Y_1^4]\right) \\ &\le \mathbb{E}[Y_1^4]\sum \frac{1}{n^4}\cdot 3n^2 \\ &= \mathbb{E}[Y_1^4]\cdot \frac{\pi^2}{2}\end{aligned}$$

which is finite, as required. Hence

$$\mathbb{P}\left(\sum_{n=1}^{\infty}\left(\frac{S_n}{n}\right)^4 < \infty\right) = 1$$

so, applying the n^{th} term test, $\left(\frac{S_n}{n}\right)^4 \to 0$ with probability 1, so taking fourth roots immediately gives the result. ∎

§10.1 Central limit theorem

Definition 10.6 (Convergence in distribution)**.** A sequence of random variables (X_n) *converges to X in distribution*, $X_n \xrightarrow{(d)} X$, if

$$F_{X_n}(x) \to F_X(x)$$

as $n \to \infty$ for all $x \in \mathbb{R}$ that are continuity points of F_X.

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Proposition 10.7

If $X_n \xrightarrow{\mathbb{P}} X$ as $n \to \infty$, then $X_n \xrightarrow{(d)} X$ as $n \to \infty$.

Proof. Let x be a continuity point of F_X, and let $\varepsilon > 0$. Observe that

$$\{X \le x - \varepsilon\} \subseteq \{X_n \le x\} \cup \{|X_n - X| > \varepsilon\}$$

and that

$$\{X_n \le x\} \subseteq \{X \le x + \varepsilon\} \cup \{|X_n - X| > \varepsilon\}$$

Taking probabilities,

$$\mathbb{P}(X \le x - \varepsilon) \le \mathbb{P}(X_n \le x) + \mathbb{P}(|X_n - X| > \varepsilon)$$

$$\implies F_X(x-\varepsilon) \leq \liminf_{n\to\infty} \mathbb{P}(X_n \leq x)$$

since the second term goes to zero by assumption. Furthermore,

$$\begin{aligned} \mathbb{P}(X_n \leq x) &\leq \mathbb{P}(X \leq x+\varepsilon) + \mathbb{P}(|X_n - X| > \varepsilon) \\ \implies \limsup_{n\to\infty} F_{X_n}(x) &\leq F_X(x+\varepsilon) \end{aligned}$$

so that

$$F_X(x-\varepsilon) \leq \liminf F_{X_n}(x) \leq \limsup F_{X_n}(x) \leq F_X(x+\varepsilon)$$

Now, taking $\varepsilon \to 0$, since x is a continuity point, we obtain

$$\liminf F_{X_n}(x) = \limsup F_{X_n}(x) = \lim_{n\to\infty} F_{X_n}(x) = F_X(x)$$

■

Proposition 10.8 (Continuity property for mgf's)

Let X_n be random variables with mgfs $m_n(\theta) = \mathbb{E}[e^{\theta X_n}]$ for $\theta \in \mathbb{R}$, and X a random variable with mgf $m(\theta) = \mathbb{E}[e^{\theta X}]$. Suppose $m(\theta) < \infty$ for some $\theta \neq 0$. If $m_n(\theta) \to m(\theta)$ as $n \to \infty$ for all $\theta \in \mathbb{R}$, then $X_n \xrightarrow{(d)} X$ as $n \to \infty$.

We will need this result for the next theorem, although we will not prove it.

Theorem 10.9 (Central limit theorem)

Let X_n be i.i.d. random variables with finite mean μ and finite variance σ^2. Let $S_n = X_1 + \cdots + X_n$, then

$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{(d)} Z$$

as $n \to \infty$, where $Z \sim \mathcal{N}(0,1)$. In other words,

$$\mathbb{P}\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \leq x\right) \longrightarrow \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}}\, dy = \Phi(x)$$

Proof. Let $Y_i = \frac{X_i - \mu}{\sigma}$, so $\mathbb{E}[Y_i] = 0$ and $\mathrm{Var}(Y_i) = 1$. It suffices, then, to show the theorem for a sequence $X_1, \dots, X_n$ i.i.d. with mean 0 and variance 1.

We will unjustifiably assume for our proof that there exists some $\delta > 0$ with $\mathbb{E}[e^{\delta X_1}] + \mathbb{E}[e^{-\delta X_1}]$ is finite.

Let $m(\theta) = \mathbb{E}[e^{\theta X_1}]$. By the continuity property of mgf's, it suffices to show that

$$\mathbb{E}\left[e^{\theta \frac{S_n}{\sqrt{n}}}\right] \longrightarrow \mathbb{E}[e^{\theta Z}] = e^{\frac{\theta^2}{2}}$$

We have

$$\mathbb{E}[e^{\theta \frac{S_n}{\sqrt{n}}}] = \left(m\left(\frac{\theta}{\sqrt{n}}\right)\right)^n$$

$$= \left(\mathbb{E}\left[\left(\frac{X_1}{\sqrt{n}} \right)^0 \right] + \mathbb{E}\left[\left(\frac{X_1}{\sqrt{n}} \right)^1 \right] \theta + \mathbb{E}\left[\left(\frac{X_1}{\sqrt{n}} \right)^2 \right] \theta^2 + \dots \right)^n$$
$$= \left(1 + \frac{\theta^2}{2n} + \left| \mathbb{E}\left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| \right)^n$$

We now claim that

$$\left| \mathbb{E}\left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| = o(|\theta|^2)$$

as $\theta \to 0$. Let $|\theta| < \delta/2$. Then

$$\begin{aligned} \left| \mathbb{E}\left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| &\leq \mathbb{E}\left[\sum_{k \geq 3} \frac{|\theta X_1|^k}{k!} \right] \\ &= \mathbb{E}\left[|\theta X_1|^3 \sum_{k \geq 0} \frac{|\theta X_1|^k}{(k+3)!} \right] \\ &\leq \mathbb{E}\left[|\theta X_1|^3 \sum_{k \geq 0} \frac{|\theta X_1|^k}{k!} \right] \\ &= \mathbb{E}\left[|\theta X_1|^3 e^{|\theta X_1|} \right] \\ &\leq \mathbb{E}\left[|\theta X_1|^3 e^{\frac{\delta |X_1|}{2}} \right] \end{aligned}$$

Observe

$$|\theta X_1|^3 = |\theta|^3 \frac{\left| \frac{\delta}{2} X_1 \right|^3}{3!} \cdot \frac{3!}{\left(\frac{\delta}{2} \right)^2} \leq C |\theta|^3 e^{\frac{\delta |X_1|}{2}}$$

where $C = 3! \cdot \frac{2^3}{\delta^3}$ (where does this come from)? Hence

$$\begin{aligned} \mathbb{E}\left[|\theta X_1|^3 e^{\frac{\delta |X_1|}{2}} \right] &\leq C |\theta|^3 \mathbb{E}\left[e^{\frac{\delta |X_1|}{2}} \cdot e^{\frac{\delta |X_1|}{2}} \right] \\ &= C |\theta|^3 \mathbb{E}\left[e^{\delta |X_1|} \right] \\ &\leq C |\theta|^3 \mathbb{E}\left[e^{\delta X_1} + e^{-\delta X_1} \right] \end{aligned}$$

so that

$$\left| \mathbb{E}\left[\sum_{k \geq 3} \frac{\theta^k}{(\sqrt{n})^k} \cdot \frac{1}{k!} X_1^k \right] \right| \leq C' \cdot |\theta|^3$$

Hence the LHS is $O(|\theta|^3)$, so $o(|\theta|^2)$ as required. Now

$$\begin{aligned} m\left(\frac{\theta}{\sqrt{n}} \right) &= 1 + \frac{\theta^2}{2n} + o\left(\frac{\theta^2}{n} \right) \\ \implies \left(m\left(\frac{\theta}{\sqrt{n}} \right) \right)^n &\to e^{\frac{\theta^2}{2}} \end{aligned}$$

as $n \to \infty$ (aren't we implicitly taking $\theta \to 0$ here? how can this be valid for any $\theta \neq 0$ then? since we take $\theta \to 0$ for little o of $|\theta|^2$ as well as $n \to \infty$). ■

Example 10.10

Suppose $S_n \sim \text{Bin}(n, p)$. We can write

$$S_n = X_1 + \cdots + X_n$$

where the X_i are i.i.d. Bernoulli random variables with parameter p. Hence, inserting the mean and variance of the Bernoulli distribution, we obtain

$$\frac{S_n - np}{\sqrt{np(1-p)}} \xrightarrow{\text{(d)}} \mathcal{N}(0, 1)$$

so that

$$S_n \approx \mathcal{N}(np, np(1-p))$$

for large n.

Example

Suppose $S_n \sim \text{Poi}(n)$, so that

$$S_n = X_1 + \cdots + X_n$$

where the X_i are i.i.d. distributed Poi(1). Subbing in the mean and variance of the Poisson distribution, we obtain

$$\frac{S_n - n}{\sqrt{n}} \xrightarrow{\text{(d)}} \mathcal{N}(0, 1)$$
$$\implies S_n \approx \mathcal{N}(n, n)$$

Lecture 24 – 18/03/26

Suppose that, in a referendum, every individual votes yes with probability p and no with probability $1 - p$, independently of one another, and that we want to estimate p by sampling.

Let X_i be 1 if the i^{th} individual sampled voted 'yes', 0 otherwise. Let $S_N = X_1 + \cdots + X_n$, and let $\hat{p}_N = \frac{S_N}{N}$, which goes to p a.s. as $n \to \infty$ by the strong law of large numbers – in particular, $S_N \sim \text{Bin}(n, p)$. For a given desired error, how large should N be?

By the CLT,

$$S_N \approx Np + \sqrt{Np(1-p)}Z$$
$$\implies \hat{p}_N \approx p + \frac{\sqrt{p(1-p)}}{\sqrt{N}}Z$$
$$\implies |\hat{p}_N - p| \approx \frac{\sqrt{p(1-p)}}{\sqrt{N}}|Z|$$

Hence, we want to solve

$$\mathbb{P}\left(\frac{\sqrt{p(1-p)}}{\sqrt{N}}|Z| \geq \varepsilon\right) \leq p'$$

We can solve this using Φ, and we can avoid worrying about not knowing what p is by simply using the fact that $\sqrt{p(1-p)} \leq \frac{1}{\sqrt{2}}$.

§11 Simulation of random variables

Suppose one can generate uniform random numbers in $[0,1]$. How can we sample some other random variable?

Suppose first F is one-to-one – then set $X = F^{-1}(U)$, so

$$\mathbb{P}(X \leq x) = \mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

If F is not one-to-one, we define its *generalised inverse*

$$G(u) = \inf\{x \in \mathbb{R} : u \leq F(x)\}$$

Lemma 11.1

$G(u) \leq x$ if and only if $u \leq F(x)$.

Proof. The reverse implication is immediate. For the forward implication, observe there exists $x_n \geq G(u)$ such that $x_n \to G(u)$ from above and $u \leq F(x_n)$. By right continuity of F, $F(x_n) \to F(G(u))$ as $n \to \infty$, so $u \leq F(G(u))$. Hence, if $G(u) \leq x$, then $u \leq F(G(u)) \leq F(x)$ as F is increasing. ■

Now, we can set $X = G(U)$, so that

$$\mathbb{P}(X \leq x) = \mathbb{P}(G(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x)$$

§11.1 Box-Muller transform

We would like to simulate X and Y independent $\mathcal{N}(0,1)$ using two independent $U[0,1]$ random variables, U and V. Theoretically, we can just apply Φ^{-1} to each of the uniform random variables, but indeed this does not have a closed form.

Consider making a complex number $X+iY$ from X and Y – we have already shown that, for this complex number, R has density $re^{-\frac{r^2}{2}}$, $\theta \sim U[0,2\pi]$, and these are independent. Now put

$$\theta = 2\pi U$$
$$R = \sqrt{-2\log V}$$

where the R expression comes from the inverse of the CDF for R. Indeed, we have

$$\mathbb{P}(R \geq r) = \mathbb{P}\left(\sqrt{-2\log V} \geq r\right) = \mathbb{P}\left(V \leq e^{-\frac{r^2}{2}}\right) = e^{-\frac{r^2}{2}}$$

so R has the correct cdf.

§11.2 Rejection sampling

Suppose we have an r.v. X with density

$$f(x) = \frac{1(x \in A)}{|A|}$$

where $A \in [0,1]^d$ has volume $|A|$. We can sample uniformly in A by sampling coordinates uniformly, and discarding samples until we end up in A.

Let $N = \min\{n : U_n \in A\}$, and $X = U_N$ – we need to show X has density f. It suffices to show that

$$\mathbb{P}(X \in B) = \frac{|B \cap A|}{|A|}$$

We have

$$\begin{aligned}
\mathbb{P}(X \in B) &= \sum_{n=1}^{\infty} \mathbb{P}(U_n \in B,\ N = n) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(U_n \in B, U_1 \notin A, \ldots, U_{n-1} \notin A, U_n \in A) \\
&= \sum_{n=1}^{\infty} \mathbb{P}(U_n \in B \cap A)\mathbb{P}(U_1 \notin A)^{n-1} \\
&= \sum_{n=1}^{\infty} |B \cap A| \, (1 - |A|)^{n-1} \\
&= \frac{|B \cap A|}{|A|}
\end{aligned}$$